System Development

Fire Performance Test Report

Test standard: BS 8414-2:2020

Test sponsor: Eurocon Building Industries FZE / Alubond Europe d.o.o

Products: Alubond (FR-A2 ACP) Rain Screen Facade System

Report number: DLR2139

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Signed and issued on behalf of Al Futtaim Element Materials Technology Dubai L.L.C				

Executive summary

This report documents the findings of the fire test for a non-loadbearing external wall system (Alubond FR-A2 Rain Screen Facade System) undertaken on 8 June 2022 in accordance with BS 8414-2:2020 and AFE method statement DMC4170 Rev.0

Al Futtaim Element Materials Technology Dubai L.L.C (AFE) performed the test, at their laboratory in Dubai, at the request of Eurocon Building Industries FZE / Alubond Europe d.o.o.

Table 1 provides a summary of the test specimen. A summary of the results is provided in Table 2.

Table 1 Test specimen

Item	Detail
Test specimen	Aluminium composite panel cladding assembly (Alubond Rain Screen Facade System) mainly comprised of: • 4 mm thick Alubond FR-A2 ACP • Alubond RFS - Galvanized steel bracket • Alubond RFS - Aluminium bracket and runner • Alubond RFS - Aluminium flat bar (panel gap reinforcement) and aluminium panel attachment clip • Siderise RVG 90/30, vertical cavity barrier • Siderise RH25, 90/30, intumescent horizontal cavity barrier • Knauf Rocksilk rainscreen slab, 100 mm thick • Dupont Tyvek Supro breather membrane • Knauf glass wool FS-Slab, 100 mm thick and 24 kg/m^3 • Knauf steel framing system (SFS) • Knauf, Type X gypsum board

Table 2 Test results

The external wall system was tested to full test duration requirements of BS 8414-2:2020 without any early termination.

Parameter	Results
T_s, start temperature	31 °C
t_s, start time	2 minutes 00 seconds after ignition of crib
Peak temperature / time at Level 1, external	924 °C at 16 minutes 10 seconds after t_s (thermocouple 3)
Peak temperature / time at Level 2, external	695 °C at 16 minutes 10 seconds after t_s (thermocouple 11)
Peak temperature / time at Level 2, mid-depth of Cavity	271 °C at 21 minutes 40 seconds after t_s (thermocouple 19)
Peak temperature / time at Level 2, mid-depth of Knauf Rocksilk Rainscreen Slab Insulation	206 °C at 20 minutes 40 seconds after t_s (thermocouple 28)
Peak temperature / time at Level 2, mid-depth of Knauf Gypsum board (front side)	104 °C at 20 minutes 50 seconds after t_s (thermocouple 36)
Peak temperature / time at Level 2, mid-depth of Knauf Glass wool Insulation	88 °C at 20 minutes 40 seconds after t_s (thermocouple 44)
Peak temperature / time at Level 2, mid-depth of Knauf Gypsum board (rear side)	81 °C at 14 minutes 00 seconds after t_s (thermocouple 52)
Peak temperature / time at Level 3, external	361 °C at 16 minutes 10 seconds after t_s (thermocouple 59)
Peak temperature / time at Level 3, mid-depth of cavity	162 °C at 22 minutes 50 seconds after t_s (thermocouple 66)

Parameter	Results
Peak temperature / time at Level 3, mid-depth of Knauf Rocksilk Rainscreen Slab Insulation	113 °C at 28 minutes 00 seconds after t_s (thermocouple 78)
Peak temperature / time at Level 3, mid-depth of Knauf Gypsum board (front side)	75 °C at 29 minutes 40 seconds after t_s (thermocouple 86)
Peak temperature / time at Level 3, mid-depth of Knauf Glass wool Insulation	68 °C at 29 minutes 30 seconds after t_s (thermocouple 94)
Peak temperature / time at Level 3, mid-depth of Knauf Gypsum board (rear side)	41 °C at 28 minutes 30 seconds after t_s (thermocouple 100)

Contents

1. Introduction

This report documents the findings of the fire test for a non-loadbearing external wall system (Alubond (FR-A2 ACP) Rain Screen Facade System) undertaken on 8 June 2022 in accordance with BS 8414-2:2020.

Al Futtaim Element Materials Technology Dubai L.L.C (AFE) performed the test at the request of the test sponsor listed in Table 3.

Table 3 Test sponsor details

Test sponsors	Address
Eurocon Building Industries FZE	Hamriyah Freezone P.O. Box 49990 Sharjah Unite Arab Emirates
Alubond Europe d.o.o | Aluminium Composite Panels	Nemanjina 130 P.O. Box 26320 Banatski Karlovac Serbia

AFE is accredited to ISO/IEC 17025:2017 by the United Kingdom Accreditation Service (UKAS), which assesses the technical competence of the laboratory, as well as its quality management systems.

Warringtonfire Testing & Certification Limited, trading as Warringtonfire and AFE are part of the Element Material Technology Group.

This test report is personal to the client, confidential, non-assignable and shall not be reproduced, except in full, without the prior, written approval of AFE and relates only to the actual specimen as tested and described herein.

The test was witnessed wholly or in part by the test witnesses listed in Table 4.

Table 4 Test witnesses

Test witness	Company
Ramachandram Gurijala	Eurocon Building Industries FZE
Rajiv Jha	Metal Plast Industries FZE
Rasheed Ahmed	Alubond Middle East FZE
Sreenivas Narayanan	Siderise Insulation
Dinesh S.	Siderise Insulation

The test was conducted by Arun Kumar of AFE.

2. Test rig

The test specimen was installed on a purpose-built test rig constructed by AFE as per BS 8414-2:2020 standard.

The test apparatus is a vertical structural steel test frame, with a main test wall and a return wall (wing) at a 90° angle to, and at one side of the main test wall. The structural steel frame was constructed with universal beam, steel hollow section, pre-formed steel channel and angles as described in BS 8414-2:2020.

The main wall was provided with a combustion chamber with an opening of 2,000 mm x 2,000 mm. The main wall had a width of 3,800 mm and the wing wall with a width of 2,050 mm. The total height of the test rig was 9,800 mm. The test rig includes 200 mm x 200 mm concrete slabs provided at floor levels.

See Figure 1 for a schematic diagram of the test rig.

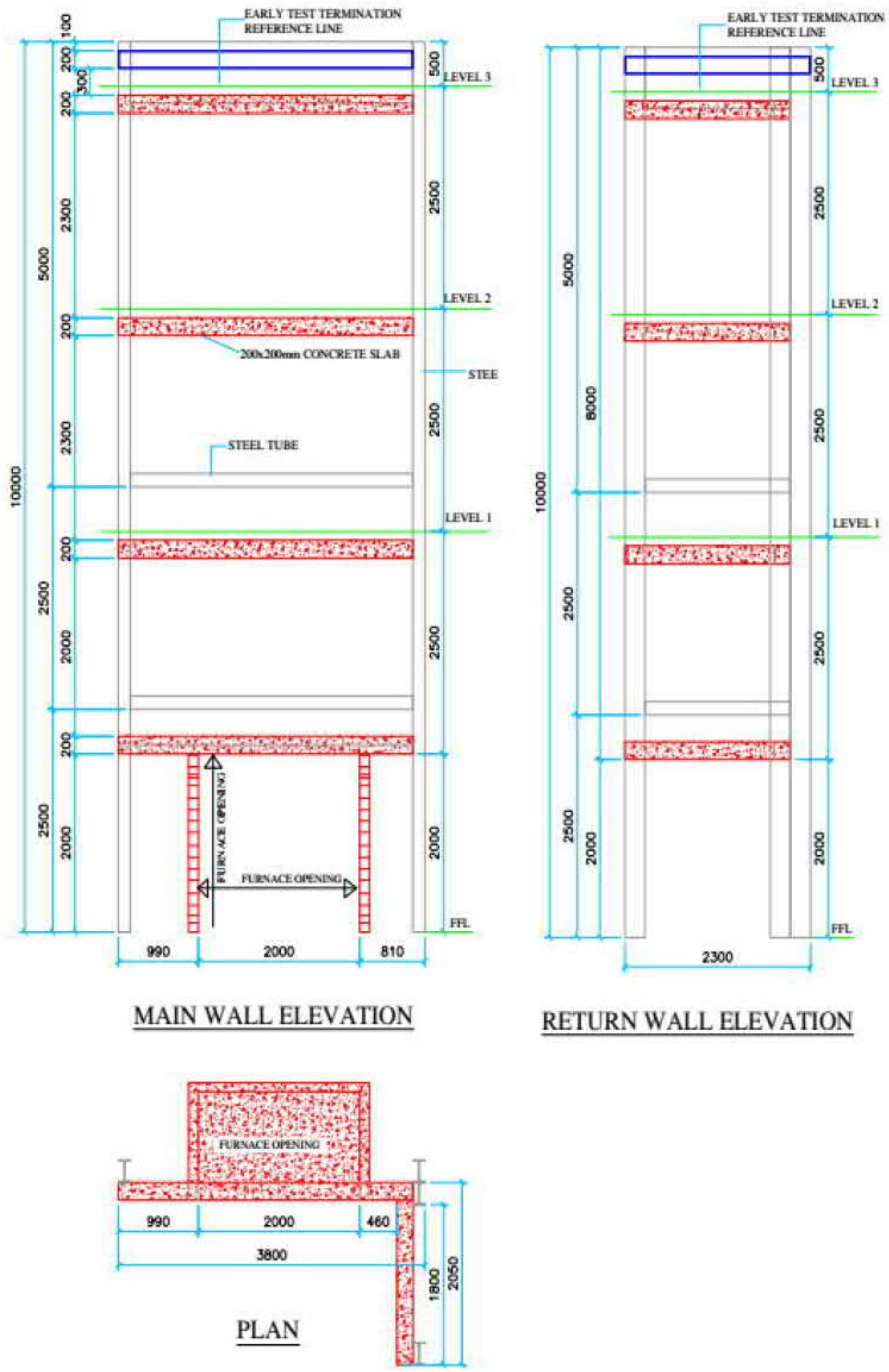


Figure 1 Schematic view of the test rig

Note: All dimensions are in mm, the drawing is not to scale

3. Test specimen

3.1 Test assembly & specimen dimensions

Table 5 provides a summary of the test assembly and Table 6 provides the dimensions of the test specimen verified by AFE after completion of the installation. A full description of the test specimen is provided in Section 3.2.

Table 5 Test assembly

Item	Detail
Test specimen	Aluminium composite panel cladding assembly (Alubond Rain Screen Facade System) mainly comprised of: • 4 mm thick Alubond FR-A2 ACP • Alubond RFS, Galvanized steel bracket • Alubond RFS, Aluminium bracket and runner • Alubond RFS, Aluminium flat bar (panel gap reinforcement) and aluminium panel attachment clip • Siderise RVG 90/30, vertical cavity barrier • Siderise RH25, 90/30, intumescent horizontal cavity barrier • Knauf Rocksilk rainscreen slab, 100 mm thick • Dupont Tyvek Supro breather membrane • Knauf glass wool FS-Slab, 100 mm thick and 24 kg/m³ • Knauf steel framing system (SFS) • Knauf, Type X gypsum board

Table 6 Dimensions of the test specimen

Item	Standard requirement	Test specimen dimensions
Width of test specimen main wall finished face	≥ 2,400 mm	3,345 mm
Width of test specimen wing wall finished face	≥ 1,200 mm	1,775 mm
Height of test specimen above the top of the combustion chamber opening	7,700 ±100 mm	7,750 mm
Height of combustion chamber opening	2,000 ±100 mm	1,998 mm
Width of combustion chamber opening	2,000 ±100 mm	1,997 mm
Distance from the combustion chamber opening to finished face of test specimen on wing wall	260 ±100 mm	320 mm

The photograph below shows an external view of the test specimen.

Photo 1 **External view of the test specimen**

3.2 Schedule of components

The test specimen was a 4 mm aluminium composite panel cladding assembly (Alubond FR A2 Rain Screen Facade System)

Table 7 describes the test specimen and lists the schedule of components. These were provided by the test sponsor and surveyed by AFE. The AFE reference number for the specimen is DC4170.

AFE was not involved in the design, procurement, installation and specification of the materials or system. Fire classification / other properties of the components detailed in Table 7 are as referred from the drawings / technical data sheet supplied by Eurocon Building Industries FZE and not verified by AFE.

All measurements were taken by AFE – unless indicated otherwise.

Detailed drawings of the test specimen are provided in Appendix D.

Table 7 Schedule of components

Item	Description	
Steel Framing System (SFS)		
1.	Component	Knauf SFS
	Size	Studs: Knauf 0.9 mm x 89 mm x 33 mm galvanised steel Tracks: Knauf 0.9 mm x 89 mm x 25 mm galvanised steel
	Installation	Tracks and studs were provided between the concrete floor slabs of the test rig. Tracks were fixed to the concrete slab with concrete drilling screws and studs were fixed to the tracks with self-drilling screws. Refer to photo 2 in Appendix C.
Internal Lining Board		
2.	Component	Knauf Type X gypsum board
	Size	15.9 mm thick, 1 layer
	Installation	Type X gypsum boards were fixed onto the steel framing system with Ø4.2 x 38mm self-drilling screws. All board joints were caulked with fibre joint tape and Knauf jointing compound.
SFS Infill Insulation		
3.	Component	Knauf glass wool FS-Slab
	Size	100 mm thick, 24 kg/m³ density
	Installation	Knauf glass wool insulation was provided between the studs and fixed to the Knauf gypsum board with 75 mm long self-adhesive insulation fastener. Refer to photo 3 in Appendix C.
Sheathing Board		
4.	Component	Knauf Type X gypsum board
	Size	15.9 mm thick, 1 layer
	Installation	Type X gypsum boards were fixed onto the steel framing system with Ø4.2 x 38 mm self-drilling screws. All board joints were caulked with fibre joint tape and Knauf jointing compound.
Breather Membrane		
5.	Component	Dupont Tyvek Supro vapour barrier
	Installation	Dupont Tyvek Supro vapour barrier was fixed to the gypsum board with pin and washer. Refer to photos 4 & 7 in Appendix C.
Brackets		
6.	Component	Alubond RFS wall bracket (ARFS-0003)
	Material	Galvanized steel
	Size	90 mm x 50 x 75 mm x 4 mm thick
	Installation	Galvanized steel brackets were fixed to Knauf gypsum board with Ø6.3 x 38mm self-drilling screw. Refer to photos 4 & 8 in Appendix C.
7.	Component	Alubond RFS Horizontal support bracket (ARFS-0002)
	Material	Aluminium
	Size	75 mm x 50 mm x 75 mm x 4 mm thick
	Installation	Aluminium brackets were fixed to galvanized steel wall brackets with M8 x 25mm bolt with washer and nut. Refer to photos 4 & 9 in Appendix C.

Item	Description	
8.	Component	Alubond RFS flashing bracket (ARFS-0010)
	Material	Galvanized steel
	Size	100 mm x 100 mm x 75 mm x 4 mm thick
	Installation	Galvanized steel brackets were provided to support GI flashing sheet around the combustion chamber and brackets were fixed to the test rig with M8 x 90 mm anchor bolt. Refer to photo 4 in Appendix C.
Cavity Barrier		
9.	Component	Siderise vertical cavity barrier, RVG 90/30
	Size	75 mm thick, 150 mm depth, 75 kg/m³ density
	Installation	Siderise vertical cavity barriers were fixed to were fixed to the Knauf gypsum board with 1 mm steel brackets and Ø4 x 25 mm self-tapping screws and plug. Refer to Figure 3 in Appendix B for the locations of cavity barriers. Refer to photos 4 & 10 in Appendix C.
10.	Component	Siderise horizontal cavity barrier, RH25,90/30
	Size	75 mm thick, 125 mm depth, 75 kg/m³ density
	Installation	Siderise horizontal cavity barriers were fixed to were fixed to the Knauf gypsum board using 1 mm steel brackets and Ø4 x 25 mm self-tapping screws and plug. Refer to Figure 3 in Appendix B for the locations of cavity barriers. Refer to photos 4 & 10 in Appendix C.
Thermal Insulation		
11.	Component	Knauf Rocksilk Rainscreen slab
	Size	100 mm thick, 55 kg/m³ density
	Installation	Knauf Rocksilk rainscreen slab insulation was fixed to the Knauf gypsum boards with Ø8 x 170 mm insulation pin. Refer to photos 5 & 11 in Appendix C.
Railings & Panel Attachment Clips		
12.	Component	Alubond RFS vertical runner (ARFS-0001)
	Material	Aluminium
	Size	40 mm x 40 mm x 3 mm thick
	Installation	Aluminium vertical runners were fixed to galvanized steel brackets with M8 x 25 mm bolt with washer and nut. Refer to photos 5, 8 & 11 in Appendix C.
13.	Component	Alubond RFS panel attachment clip (ARFS-0012)
	Material	Aluminium
	Size	80 mm long, 3 mm thick nominal
	Installation	Aluminium panel attachment clips were fixed to the aluminium runner with Ø4.2 x 19 mm self-tapping screws. Refer to photos 5 & 12 in Appendix C.
14.	Component	Alubond RFS panel perimeter strip (ARFS-0013)
	Material	Aluminium
	Size	46 mm long, 2 mm thick nominal
	Installation	Panel perimeter strips were fixed to the aluminium composite panel with ss rivets. Refer to photo 16 in Appendix C.

Item	Description	
Cladding Panel		
15.	Component	Alubond aluminium composite panel
	Alloy	Aluminium 3003
	Size	4 mm thick
	Exterior coating	PVDF coated aluminium sheet on the exterior side
	Fire classification	FR-A2
	Installation	Aluminium composite panels were fixed to the panel attachment clip via perimeter strip. 3mm aluminium angle were provided as stiffener at the cladding panel folding and fixed with SS rivets. Air cavity – 41mm Approximately 16 mm joints were provided between the panels. Refer to photos 6, 14 & 15 in Appendix C.
Panel Gap Reinforcement		
16.	Component	Alubond RFS flat bar (ARFS-0004)
	Material	Aluminium
	Size	46 mm wide, 4 mm thick
	Installation	Aluminium flat bars were provided at the panel joints via sliding on to the aluminium panel attachment clip. Refer to photo 13 in Appendix C.
Flashing		
17.	Component	Aluminium cover cap
	Size	30 mm x 147 mm x 30 mm x 1 mm thick
	Installation	Aluminium cover caps were provided at the top termination and fixed to the steel box with Ø5 x 35 mm self-drilling screws and fixed to the ACP with Ø4.2 x 19 mm self-tapping screws.
18.	Component	Galvanized steel flashing
	Size	23 mm x 350 mm x 3 mm thick
	Installation	Galvanized steel flashings were fixed around the combustion chamber with M8 x 90 mm anchor bolt with washer and nut and fixed to the ACP with Ø4.8 x 25 mm self-drilling screws. Refer to photo 17 in Appendix C.

3.3 Installation details

Table 8 lists the installation details for the test specimen.

Table 8 Installation details

Item	Detail
Start date for construction of the test specimen	10 May 2022
Completion date for construction of the test specimen	3 June 2022
External wall system constructed by	The test sponsor.
Ambient temperature range during the construction	26 – 38 °C
Pre-test conditioning period specified by test sponsor	Not specified
Pre-test conditioning dates	03 to 07 June 2021
Ambient temperature range during the conditioning	25 – 38 °C

4. Test procedure

Table 9 details the test procedure for this fire test.

Table 9 Test procedure

Item	Detail	
Statement of compliance	The test was performed in accordance with the requirements of BS 8414-2:2020 for a non-loadbearing external cladding system fixed to and supported by a structural steel frame.	
Variations	None.	
Environmental conditions at the start of the test	Start of the test	31 °C
	Wind speed	0.2 ms^{-1}
Ignition source	Crib material	Softwood (Pinus Silvestris)
	Moisture content	13 %
	Density	0.62 kg/dm^3
Sampling / specimen selection	The laboratory was not involved in sampling or selecting the test specimen for the fire test.	
Test duration	60 minutes	
Instrumentation and equipment	The instrumentation was provided in accordance with BS 8414-2:2020 as follows: • All exposed and cavity temperatures were measured by mineral insulated metal sheathed (MIMS) Type K thermocouples with an overall diameter of 1.5 mm with the measuring junction insulated from the sheath. • The thermocouple positions are shown in Figures 3 & 4 in Appendix B. • The wind speed was measured by an anemometer at Level 2, 1,000 mm forward from the centre line of the combustion chamber. • Timber crib moisture was measured by a moisture meter.	

5. Test measurements and results

Table 11 shows the peak temperatures the test specimen achieved as listed in BS 8414-2:2020.

The temperature measurements for the test specimen are included in Appendix B.

Table 12 in Appendix A includes observations of any significant behaviour of the specimen and details the occurrence of the various performance criteria specified in BS 8414-2:2020 and summarizes the observations made after the completion of the test.

Photographs of the specimen are included in Appendix C.

The performance criteria of the test were based on BR 135:2013 Annex B – Fire performance of external thermal insulation for walls of multistorey buildings, as requested by Eurocon Building Industries FZE / Alubond Europe d.o.o, and as follows:

External fire spread:

Failure due to external fire spread is deemed to have occurred if the temperature rise above T_s of any of the external thermocouples at level 2 exceeds 600 °C, for a period of at least 30 seconds, within 15 minutes of the start time, t_s.

Internal fire spread:

Failure due to internal fire spread is deemed to have occurred if the temperature rise above T_s of any of the internal thermocouples at level 2 exceeds 600 °C, for a period of at least 30 seconds, within 15 minutes of the start time, t_s.

Table 10 Key for test results

Parameter	Description
Start temperature, T_s	Mean temperature of the thermocouples at Level 1, five minutes prior to ignition of the heat source.
Start time, t_s	Time when the temperature recorded by any external thermocouple at Level 1 ≥ 200 °C above T_s and remains above this value for at least 30 seconds.
Level 1 height	2,500 mm above the top of the combustion chamber opening.
Level 2 height	5,000 mm above the top of the combustion chamber opening.
Level 3 height	7,500 mm above the top of the combustion chamber opening.

Table 11 Test results

Parameter	Results
T_s, start temperature	31 °C
t_s, start time	2 minutes 00 seconds after ignition of crib
Peak temperature / time at Level 1, external	924 °C at 16 minutes 10 seconds after t_s (thermocouple 3)
Peak temperature / time at Level 2, external	695 °C at 16 minutes 10 seconds after t_s (thermocouple 11)
Peak temperature / time at Level 2, mid-depth of cavity	271 °C at 21 minutes 40 seconds after t_s (thermocouple 19)
Peak temperature / time at Level 2, mid-depth of Knauf Rocksilk rainscreen slab insulation	206 °C at 20 minutes 40 seconds after t_s (thermocouple 28)
Peak temperature / time at Level 2, mid-depth of Knauf gypsum board (front side)	104 °C at 20 minutes 50 seconds after t_s (thermocouple 36)
Peak temperature / time at Level 2, mid-depth of Knauf glass wool insulation	88 °C at 20 minutes 40 seconds after t_s (thermocouple 44)
Peak temperature / time at Level 2, mid-depth of Knauf gypsum board (rear side)	81 °C at 14 minutes 00 seconds after t_s (thermocouple 52)
Peak temperature / time at Level 3, external	361 °C at 16 minutes 10 seconds after t_s (thermocouple 59)
Peak temperature / time at Level 3, mid-depth of cavity	162 °C at 22 minutes 50 seconds after t_s (thermocouple 66)
Peak temperature / time at Level 3, mid-depth of Knauf Rocksilk rainscreen slab insulation	113 °C at 28 minutes 00 seconds after t_s (thermocouple 78)
Peak temperature / time at Level 3, mid-depth of Knauf gypsum board (front side)	75 °C at 29 minutes 40 seconds after t_s (thermocouple 86)
Peak temperature / time at Level 3, mid-depth of Knauf glass wool Insulation	68 °C at 29 minutes 30 seconds after t_s (thermocouple 94)
Peak temperature / time at Level 3, mid-depth of Knauf gypsum board (rear side)	41 °C at 28 minutes 30 seconds after t_s (thermocouple 100)

Based on the above test results, the test specimen was found to be compliant with the requirements of BR 135:2013 Annex B.

The above results are valid only for the tested sample as received, detailed, and constructed as per the drawings with any marked variations as attached in Appendix D of this report, and the conditions under which the tests were conducted.

6. Application of test results

6.1 Test limitations

The results of these fire tests may be used to directly assess fire hazard, but it should be recognised that a single test method will not provide a full assessment of fire hazard under all fire conditions.

These results only relate to the behaviour of the specimen of the element of construction under the conditions of the test. They are not intended to be the sole criteria for assessing the potential fire performance of the element in use, and they do not necessarily reflect the actual behaviour in fires.

6.2 Variations from the tested specimen

This report details methods of construction, the test conditions and the results obtained when the specific element of construction described here was tested following the procedure outlined in BS 8414-2:2020.

Any significant variation with respect to size, construction details, loads, stresses, edge, or end conditions, other than that allowed under the field of direct application in the relevant test method, is not covered by this report.

6.3 Uncertainty of measurement

The uncertainty of measurement of the test result is not reported due to the nature of fire testing and the consequent difficulty in quantifying the uncertainty of measurements obtained from the test.

Appendix A Test observations

A.1 Visual observations

Table 12 shows the observations of any significant behaviour of the specimen during the test.

Table 12 Test observations

Time from ignition Min	Sec	Observation
00	00	Ignition of crib.
00	39	Flame tip reached above the combustion chamber.
01	12	Flame tip reached 2 m above the combustion chamber.
01	54	Flame tip reached 3 m above the combustion chamber.
03	51	Coating of mid-panels up to 1 m above the combustion chamber was observed to be peeling off. Refer to photo 18 in Appendix C.
04	16	Coating of panels up to 2 m above the combustion chamber was observed to be peeling off. Refer to photo 19 in Appendix C.
05	01	Mid-panels up to 2 m above the combustion chamber started to deform.
06	30	Flame tip observed at 4 m above the combustion chamber.
06	32	Coating of mid-panels at level 1 was observed to be peeling off.
07	15	Aluminium skin of the mid panel at 1 m started to melt.
08	26	Coating of mid-panels at 3 m above the combustion chamber was observed to be peeling off. Refer to photo 20 in Appendix C.
09	07	Aluminium skin of the mid panels at 2 m above the combustion chamber started to melt.
09	43	Coating of wing wall corner panels at 1 m was observed to be peeling off.
11	01	Mid-panels up to 1 m above the combustion chamber were partially consumed and insulation exposed. Refer to photo 21 in Appendix C.
11	48	Coating of mid panels at 4 m above the combustion chamber was observed to be peeling off.
12	17	Sustained flames on the mid-panels between 1 to 2 m above the combustion chamber.
13	07	Discoloration on wing wall corner panels up to 2 m above the combustion chamber.
13	26	Mid-panels between 1 to 2 m above the combustion chamber were partially consumed and insulation exposed. Refer to photo 22 in Appendix C.
14	11	Mid-panels between 2 to 3 m above the combustion chamber were started to melt.
14	41	Coating of mid-panels at 5 m above the combustion chamber was observed to be peeling off. Refer to photo 23 in Appendix C.
15	44	Debris started to fall off.
16	01	Flaming debris from mid panel at 1 m above the combustion chamber fell off. Refer to photo 24 in Appendix C.
16	03	Approximately 90% of mid-panels up to 2 m above the combustion chamber were consumed.
16	20	Sustained flames on the mid-panels at 2 m above the combustion chamber.
16	31	Debris of core from mid-panels between 1 to 2 m above the combustion chamber fell off. Refer to photo 25 in Appendix C.
18	00	Mid-panels between 3 to 4 m above the combustion chamber started to melt.
18	23	Debris of mid panels at 2 m above the combustion chamber fell off.
18	41	Flaming debris observed on the floor.

Time from ignition		Observation
Min	Sec	
19	16	Flaming debris fell off the main wall.
19	46	Self-sustained flames on mid panel at level 1. Refer to photo 26 in Appendix C.
21	12	Railings behind the mid-panels up to level 1 were partially consumed.
21	59	Debris of mid-panel at 3 m above the combustion chamber fell off. Refer to photo 27 in Appendix C.
24	18	Mid-panels at 3 m above the combustion chamber were partially consumed and insulation exposed. Refer to photo 28 in Appendix C.
30	00	The heat source was extinguished, observations continued for another 30 minutes.
31	25	All visible flames ceased.
45	45	No significant changes on the test specimen.
60	00	Test was terminated.

A.2 Post-test observations

Table 13 below summarises the observations made after the test was completed.

Table 13 Post-test observations

Component	Observations
Alubond FR-A2 ACP	Main wall: Mid-panels up to level 1 were partially consumed and remaining areas were discoloured and distorted. Mid-panels between level 1 to level 2 were discoloured No significant changes on the remaining panels on the main wall. Wing wall: Wing wall panels up to level 1 were discoloured. No significant changes on the remaining panels on the wing wall. Refer to photo 29 in Appendix C.
Siderise cavity barriers	**Horizontal intumescent cavity barrier:** Main wall: The 1^{st} horizontal cavity barrier at 140 mm above the combustion chamber was activated Intumescent layer was partially present on the activated section of the cavity barrier. Minor material loss observed. The 2^{nd} horizontal cavity barrier at 2,240 mm above the combustion chamber was activated Intumescent layer was partially present on the activated sections of the cavity barrier. Material loss of mineral wool insulation of the cavity barrier was observed. The 3^{rd} horizontal cavity barrier at 4,805 mm above the combustion chamber was activated. Expanded intumescent layer was present on the activated sections of the cavity barrier. Cavity barrier was intact, and no material loss of mineral wool insulation of the cavity barrier was observed. The 4^{th} horizontal cavity barrier at 7,300 mm above the combustion chamber was not activated. Cavity barrier was intact, and no material loss of mineral wool insulation of the cavity barrier was observed. Wing wall: The 1^{st} horizontal cavity barrier at 2,140 mm above the ground floor level combustion chamber was activated. Expanded intumescent layer was present on the activated sections of the cavity barrier. Cavity barrier was intact, and no material loss of mineral wool insulation of the cavity barrier was observed. The 2^{nd} horizontal cavity barrier at 4,240 mm above the ground level was activated. Expanded intumescent layer was present on the activated section of the cavity barrier. Cavity barrier was intact, and no material loss of mineral wool insulation of the cavity barrier was observed. The 3^{rd} horizontal cavity barrier at 6805 mm above the ground level was not activated. Cavity barrier was intact, and no material loss of mineral wool insulation of the cavity barrier was observed. The 4^{th} horizontal cavity barrier at 9300 mm above the ground level was not activated. Cavity barrier was intact, and no material loss of mineral wool insulation of the cavity barrier was observed. Refer to photos 30, 34 to 36 in Appendix C.
	Vertical cavity barrier: Discolouration and distortion were observed on the on the vertical cavity barriers provided on the main wall and wing wall. Refer to photos 30, 34 to 36 in Appendix C.
Knauf Rocksilk rainscreen slab insulation	Main wall: Knauf rainscreen slab insulation located behind mid-panels up to 4 m above the combustion chamber were discoloured. All other insulation on the main wall was intact and no significant changes were observed. Wing wall: No significant changes were observed. Refer to photos 30. 34 & 35 in Appendix C.

Component	Observations
Alubond RFS Railings and panel attachment clips	Main wall: Railings and panel attachment clips located behind the mid-panels up to level 1 were melted. Railings and panel attachment clips located behind the mid-panels between level 1 and level 2 were discoloured. All other railings and panel attachment clips on the main wall were intact and no damage was observed. Wing wall: All railings and panel attachment clips on the wing wall were intact and no significant changes were observed. Refer to photos 30, 34 to 36 in Appendix C.
Alubond RFS brackets	Galvanized steel brackets All galvanized brackets on the main wall and wing wall were intact and no significant changes were observed. Aluminium horizontal support brackets: Minor discoloration was observed on the aluminium brackets located behind mid panels up to level 1. All other aluminium brackets on the main wall and wing wall were intact and no significant changes were observed. Refer to photo 31 in Appendix C.
Dupont Tyvek Supro membrane	Material loss and discoloration were observed on Tyvek membrane located behind the mid-panels up to 2 m above the combustion chamber. Remaining Tyvek membrane on the main wall and wing wall were intact and no significant changes were observed. Refer to photo 31 in Appendix C.
Knauf glass wool FS-Slab	No significant changes were observed Refer to photo 33 in Appendix C.
Knauf type X Gypsum board and SFS	No significant changes were observed. Refer to photos 32 & 33 in Appendix C.

Refer to Figure 3 in Appendix B for panel numbering.

Approximately 3.5 m^2 of the external visible area was completely consumed.

Approximately 12 m^2 of the external visible area was discoloured.

See Figure 2 for more details.

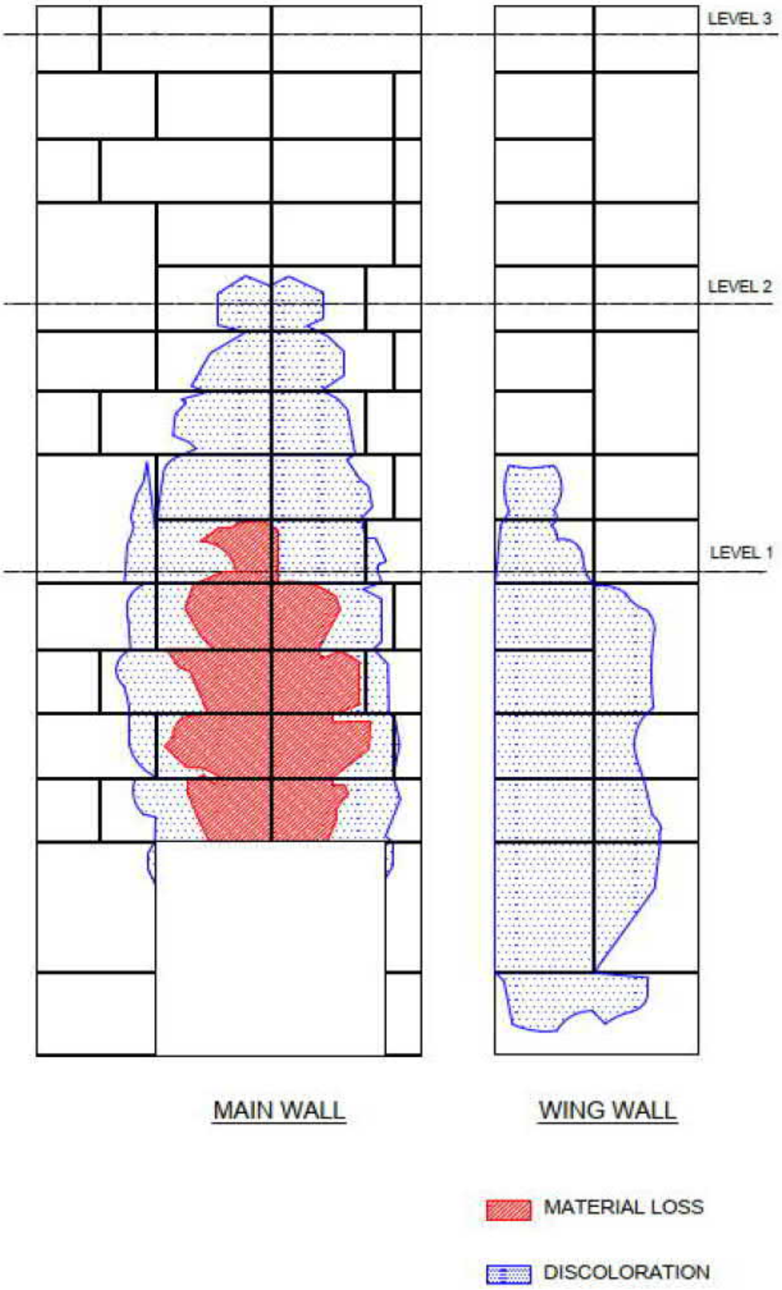


Figure 2 Area map showing the condition of the Alubond FR-A2 ACP cladding panels after the test

Appendix B Test data

B.1 Thermocouple, cavity barrier locations and panel numbering

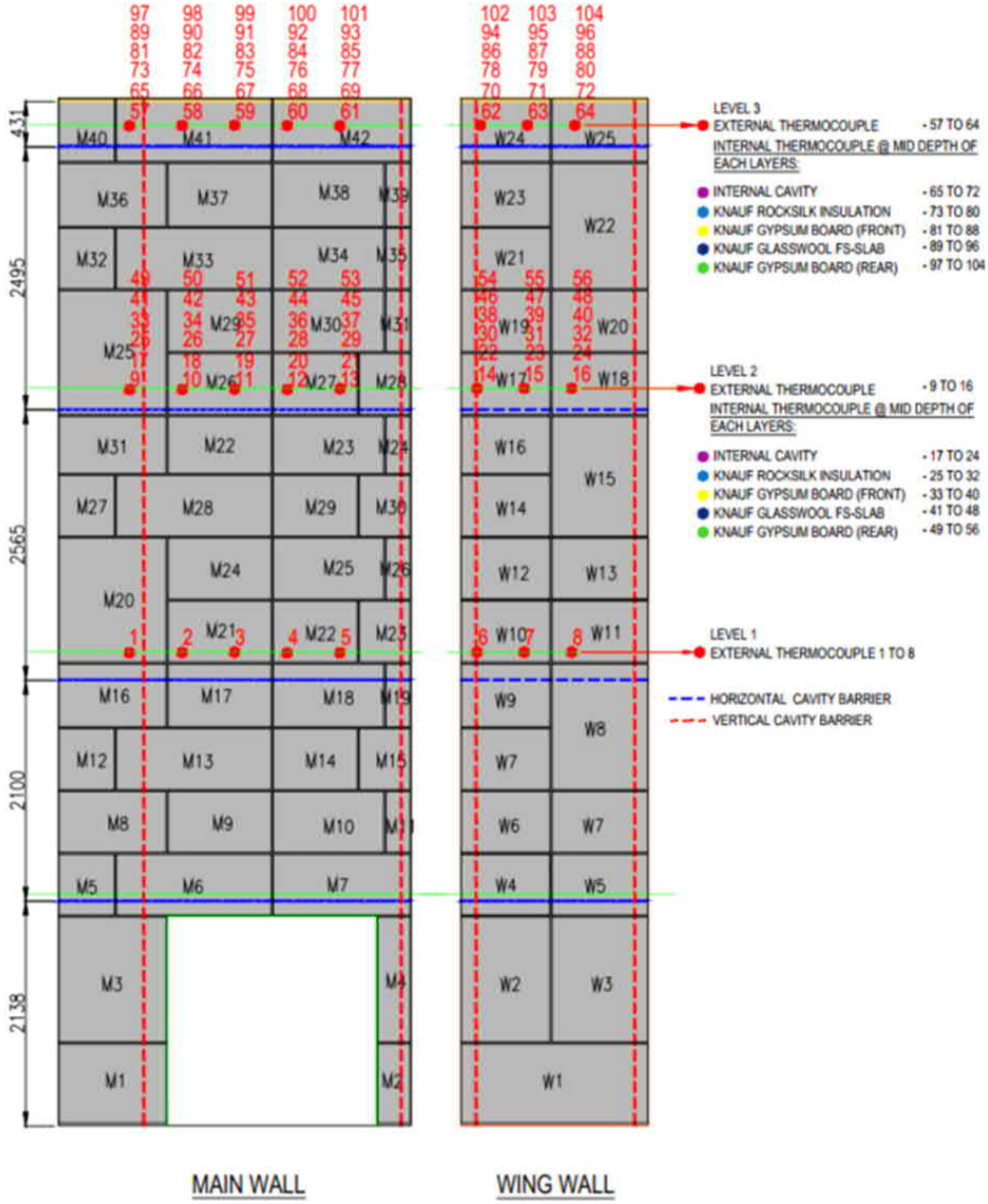
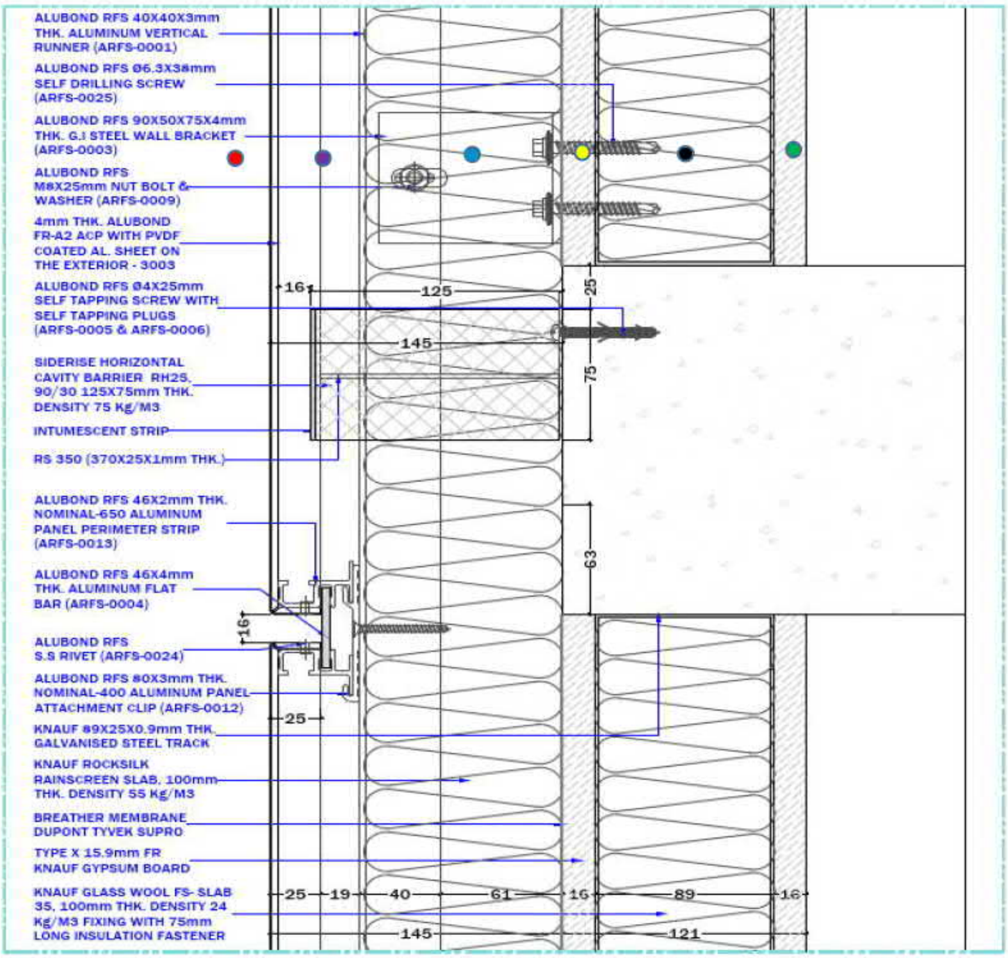


Figure 3 Thermocouple , cavity barrier locations and panel numbering

B.2 Level 2 and 3 section drawing showing the thermocouple locations



●	External thermocouples	- 9 to 16 & 57 to 64
●	Internal Cavity	- 17 to 24 & 65 to 72
●	Knauf Rocksilk insulation	- 25 to 32 & 73 to 80
●	Knauf Gypsum Board (front)	- 33 to 40 & 81 to 88
●	Knauf Glasswool FS-Slab	- 41 to 48 & 89 to 96
●	Knauf Gypsum Board (rear)	- 49 to 56 & 97 to 104

Figure 4 Levels 2 and 3 section drawing showing the thermocouple locations

B.3 Specimen temperatures

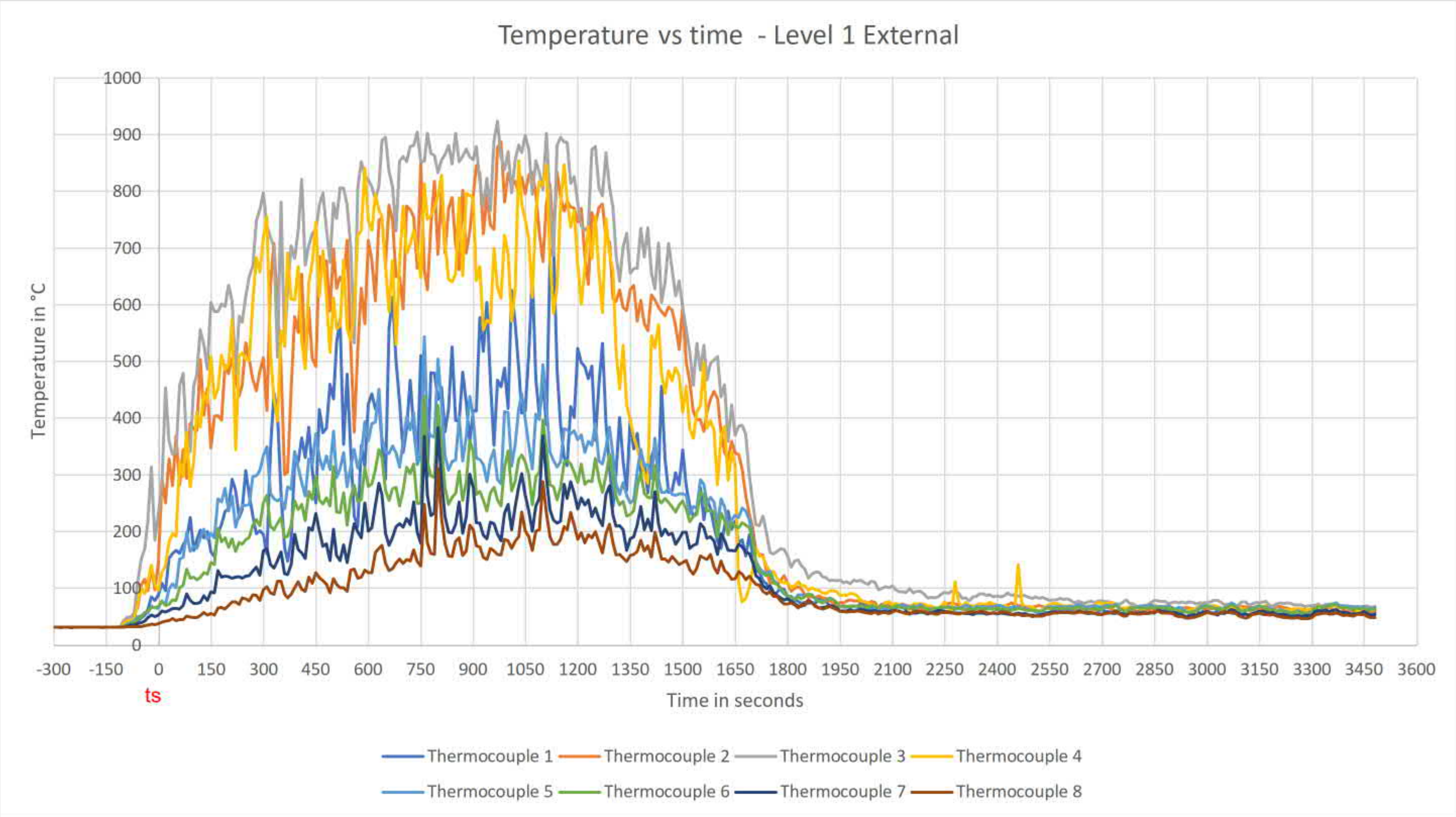
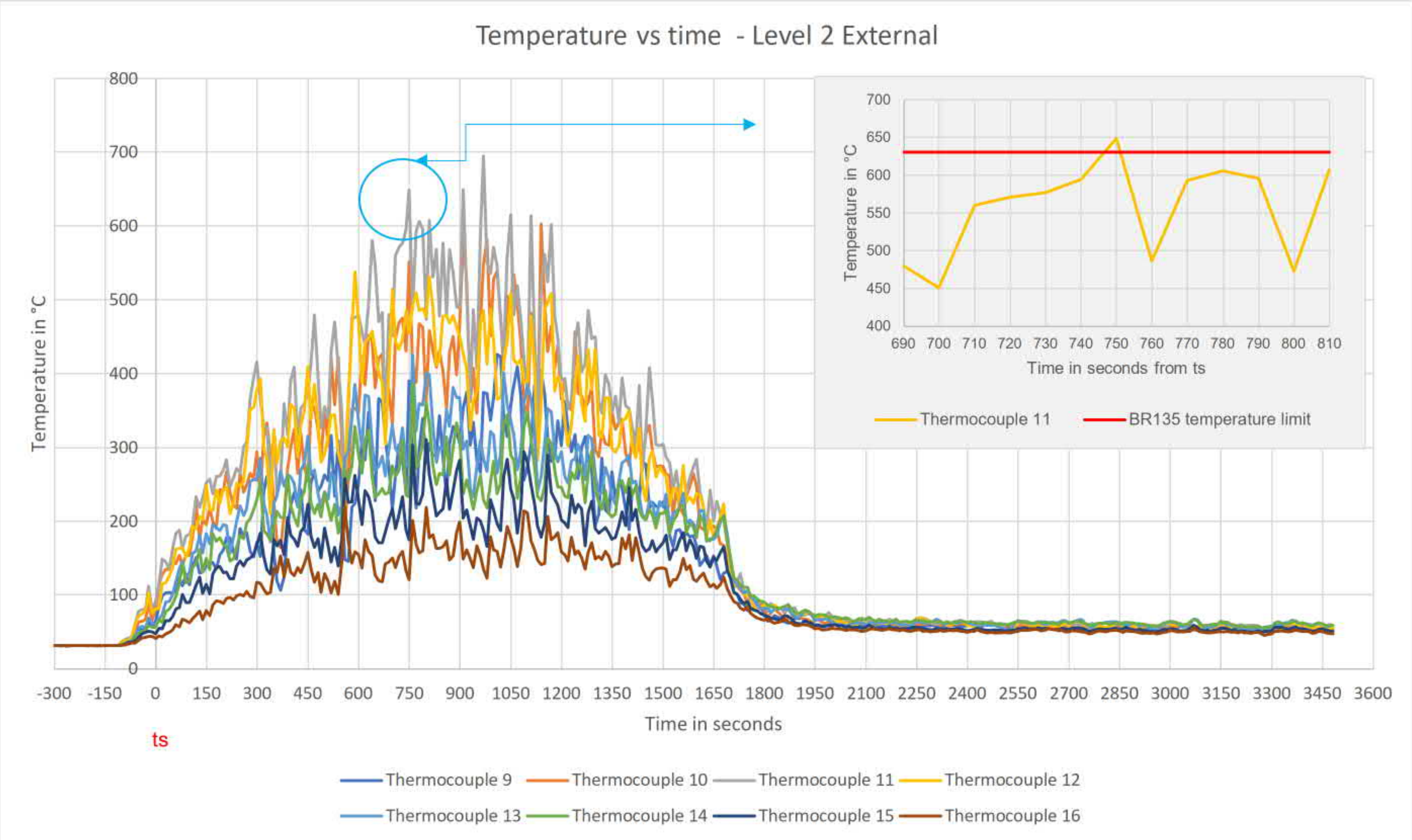


Figure 5 **Level 1, external – temperature vs time**



Note: Within 15 minutes of the start time (ts), thermocouple 11 recorded a temperature of 600ºC above Ts (631 °C), sustained for a period of less than 10 seconds.

Figure 6 Level 2, external – temperature vs time

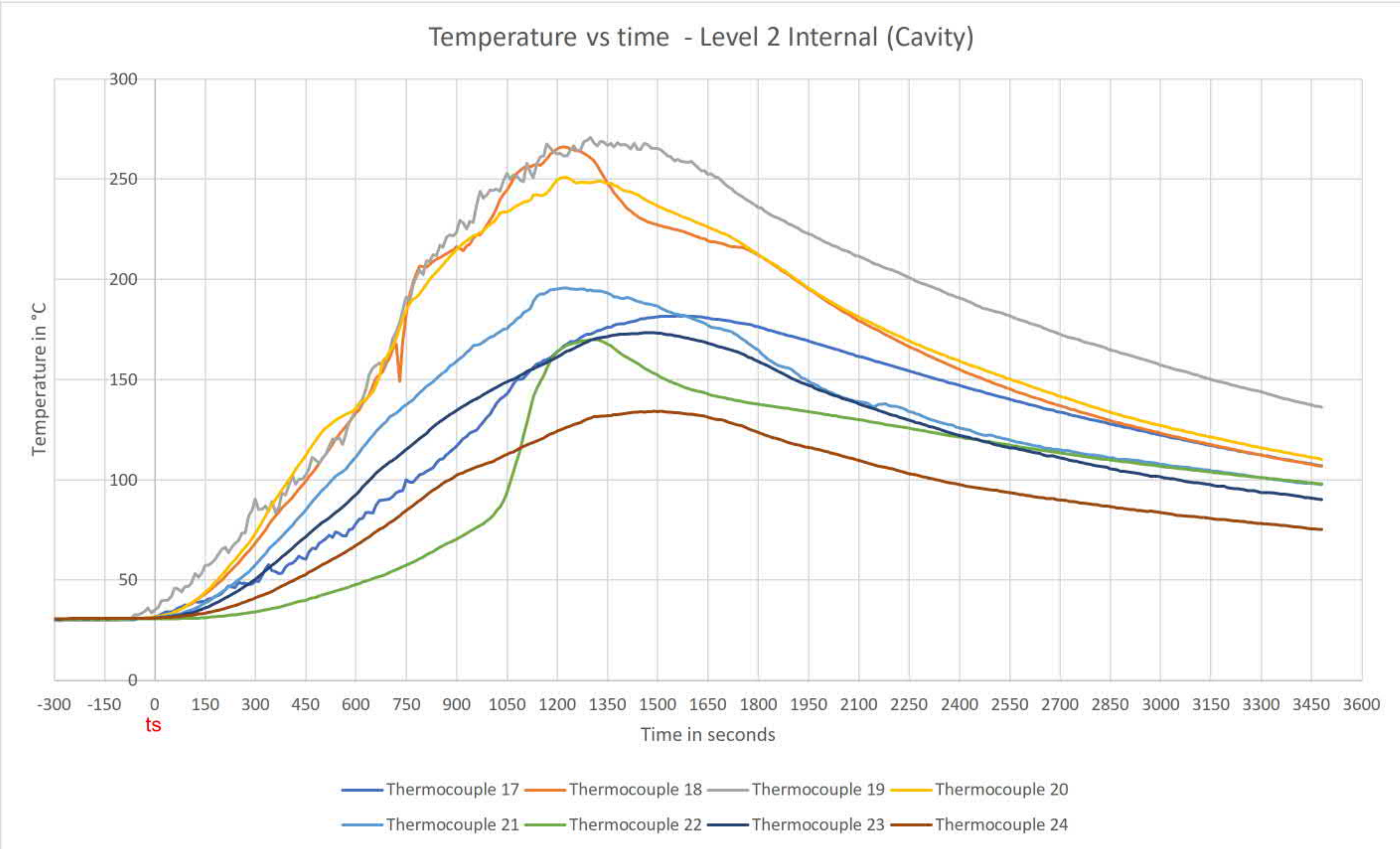


Figure 7 Level 2, internal cavity – temperature vs time

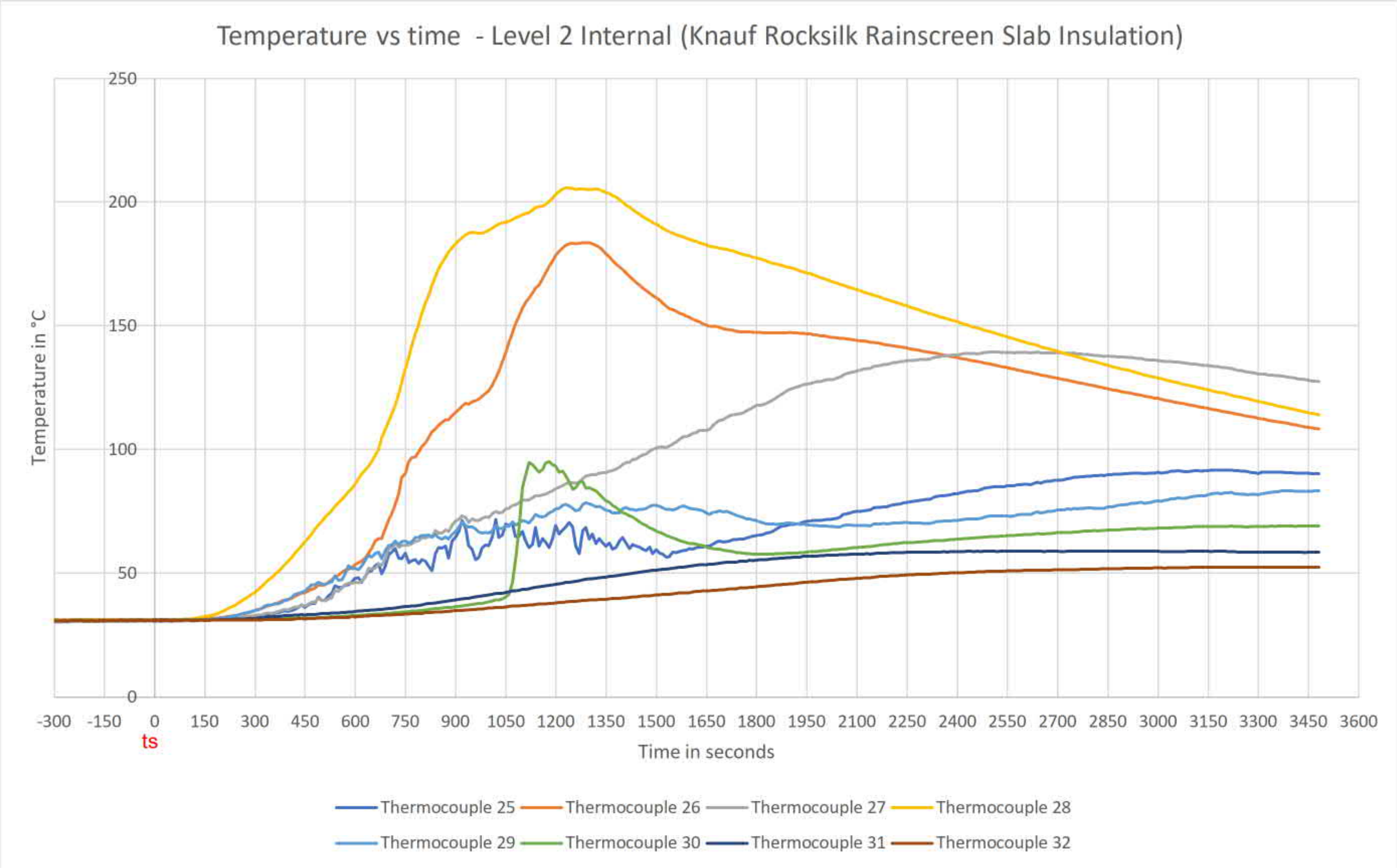


Figure 8 **Level 2, internal, Knauf Rocksilk rainscreen insulation – temperature vs time**

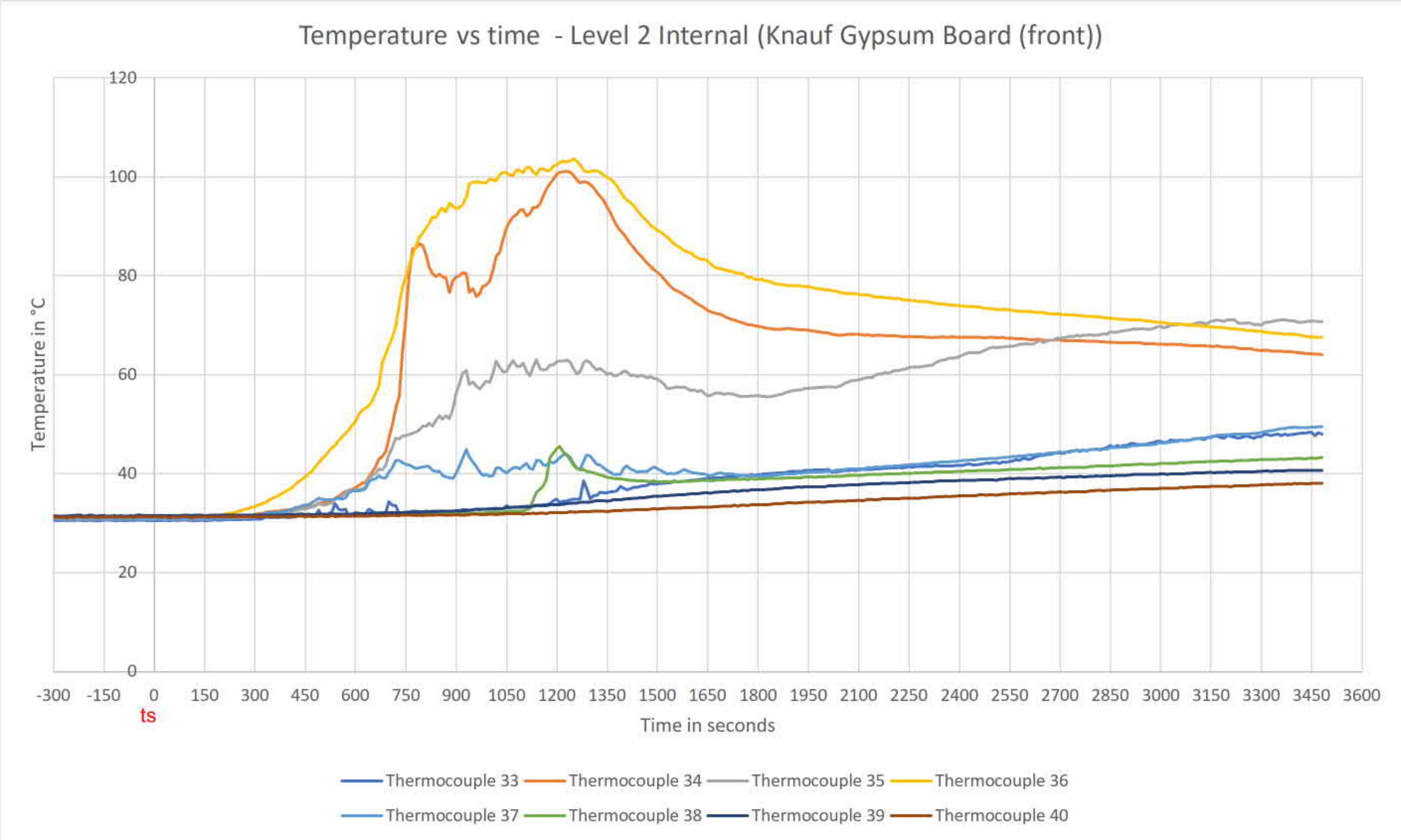


Figure 9 Level 2, internal, Knauf gypsum board (front) – temperature vs time

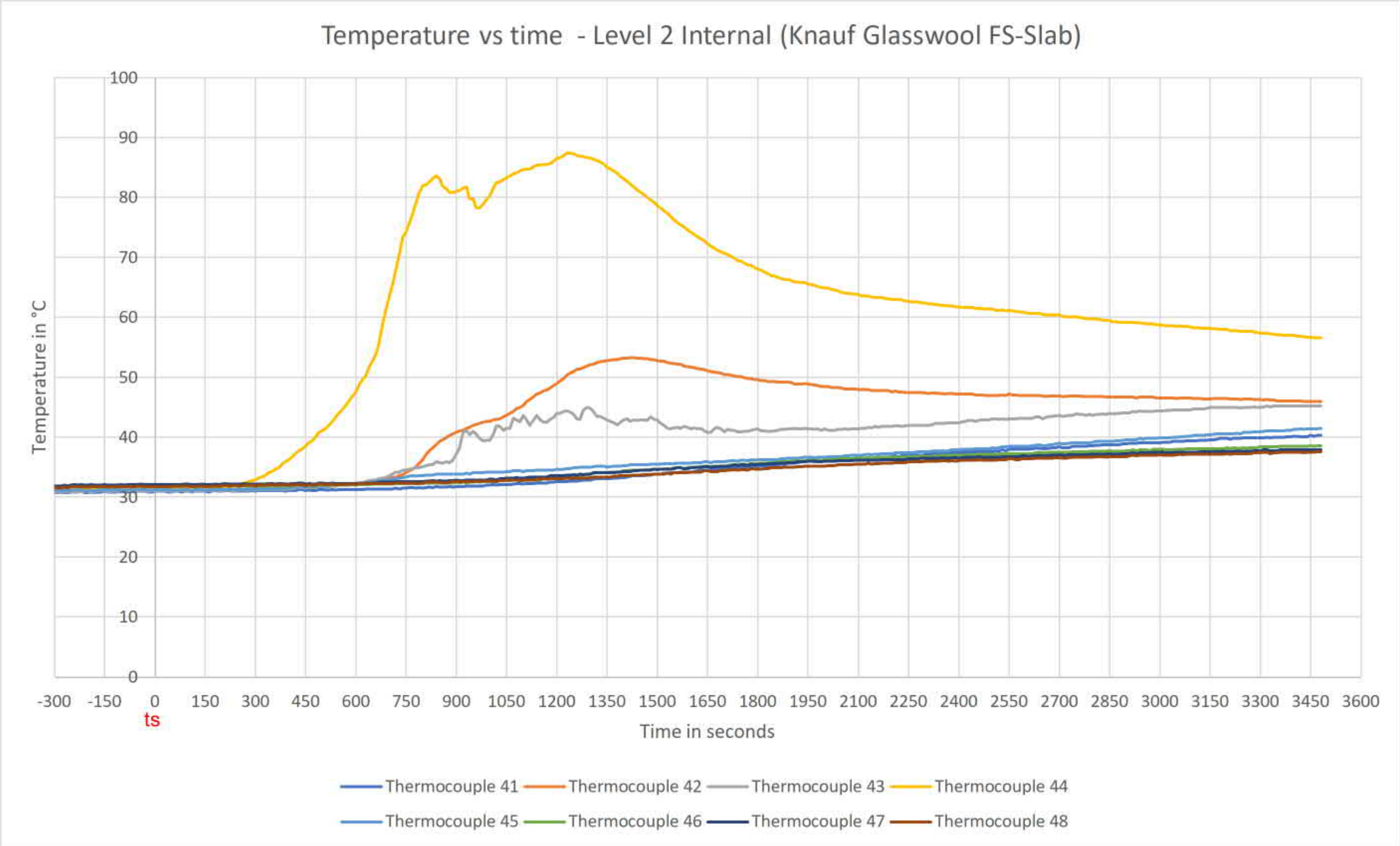


Figure 10 Level 2, internal, Knauf glasswool FS-Slab – temperature vs time

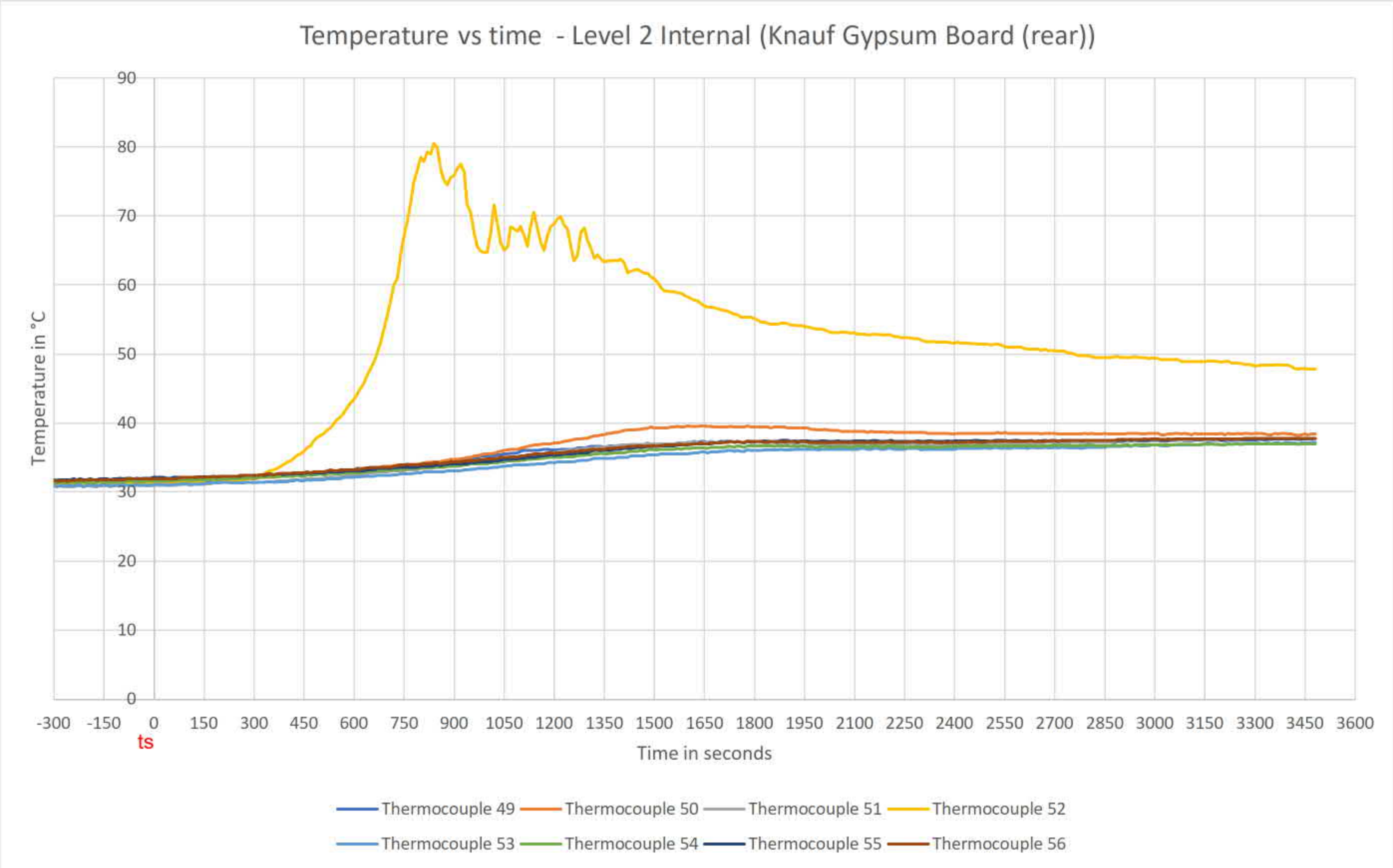


Figure 11 Level 2, internal, Knauf gypsum board (rear) – temperature vs time

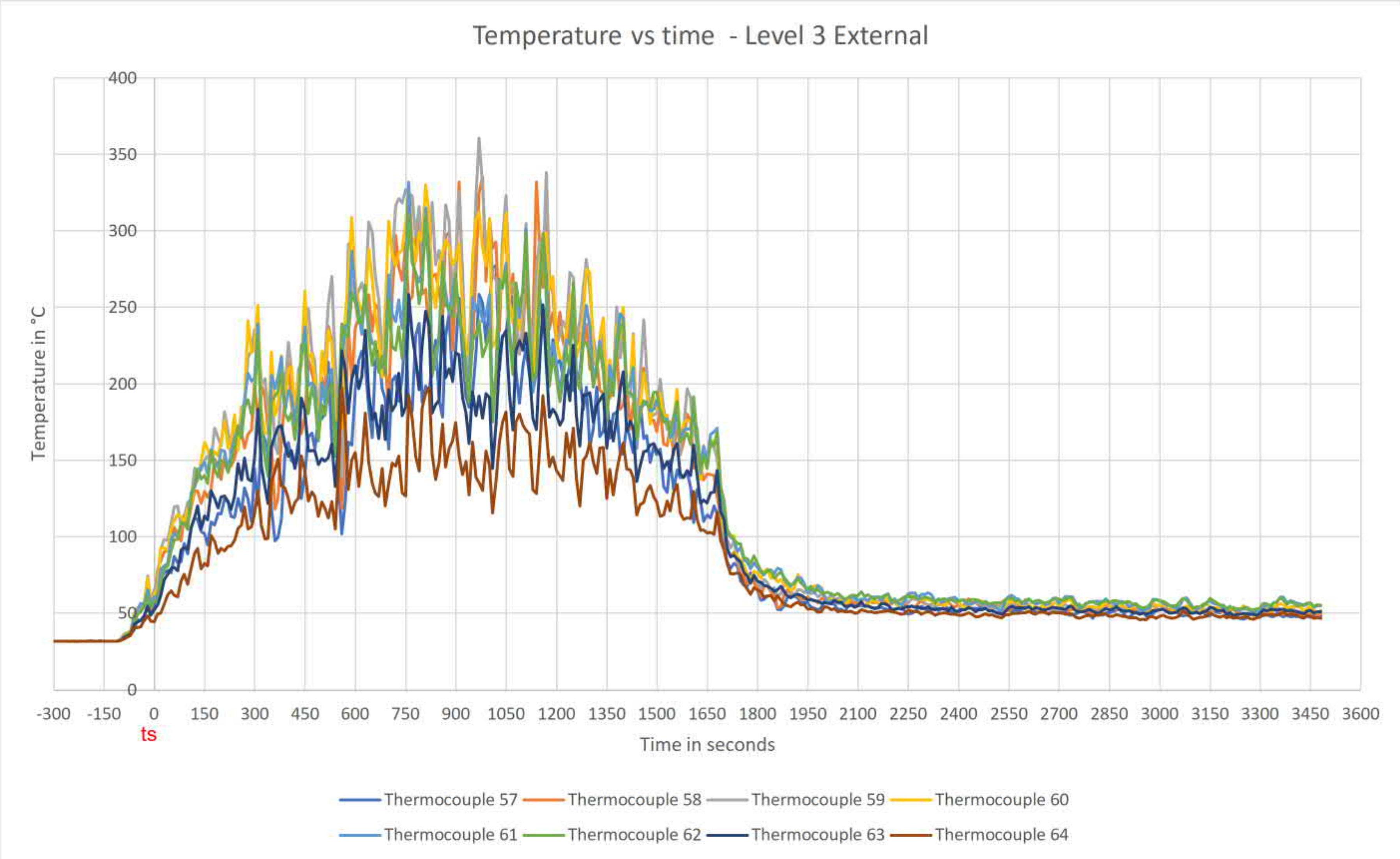


Figure 12 Level 3, external – temperature vs time

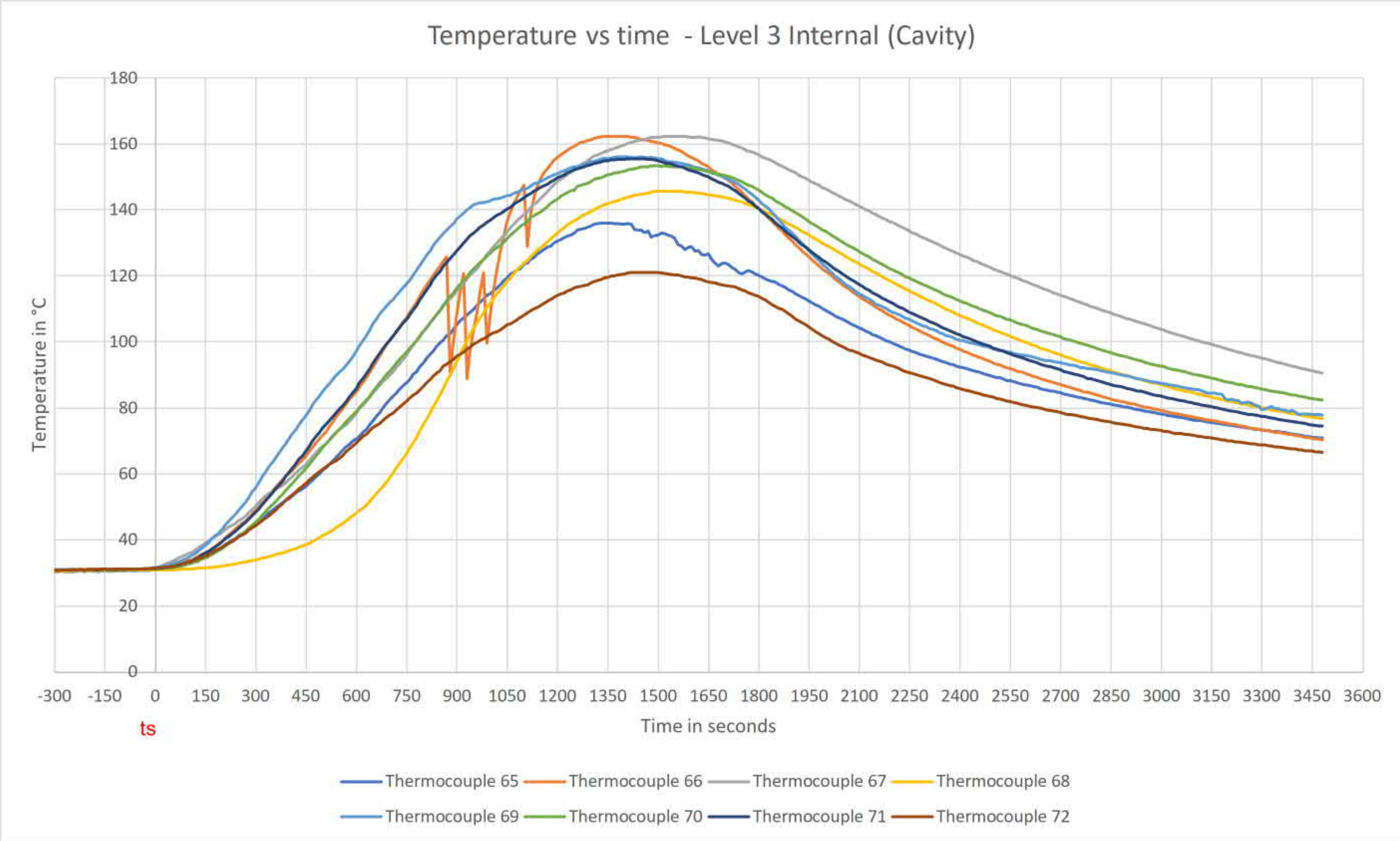


Figure 13 Level 3, internal, cavity – temperature vs time

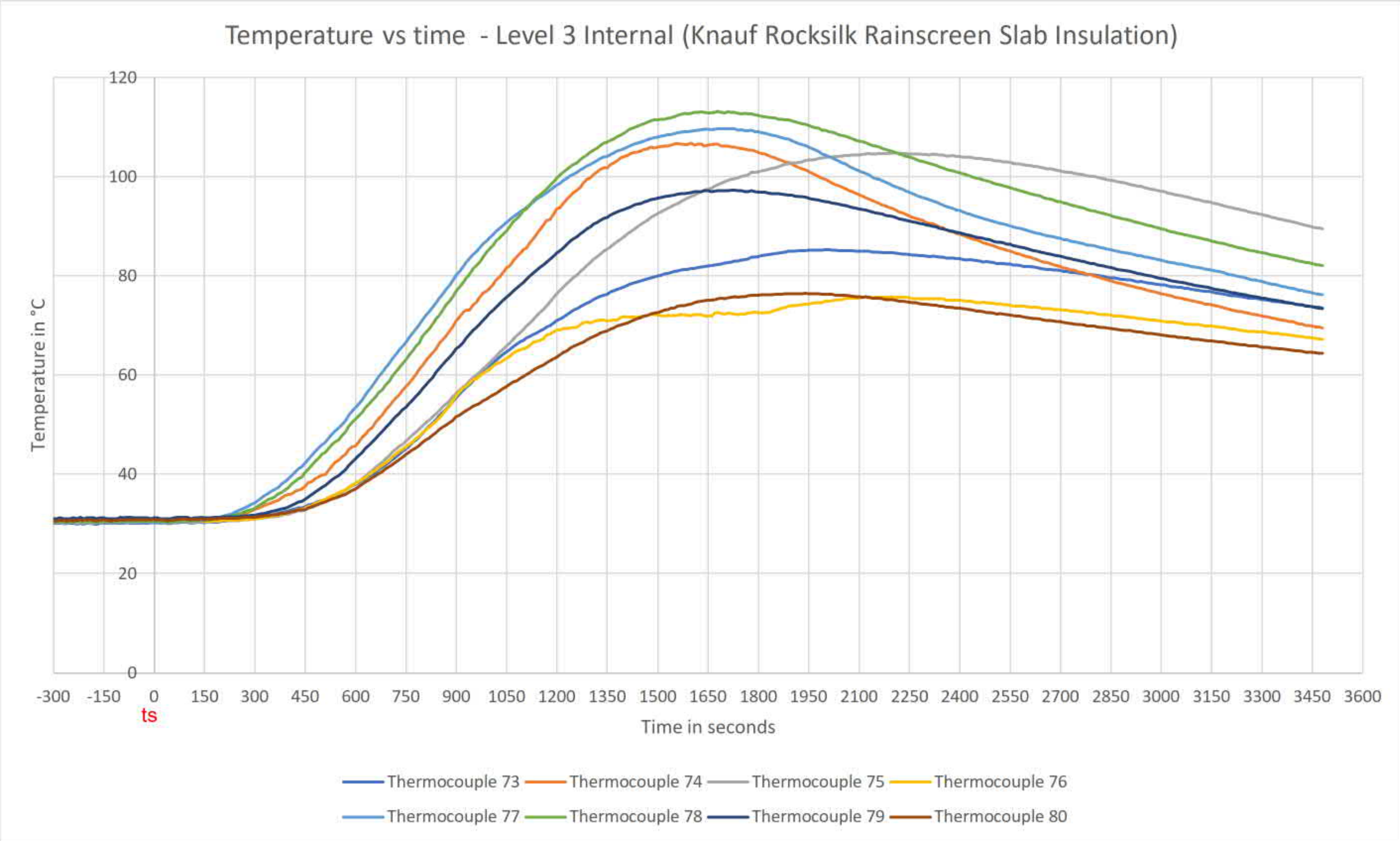


Figure 14 Level 3, internal, Knauf Rocksilk rainscreen Insulation – temperature vs time

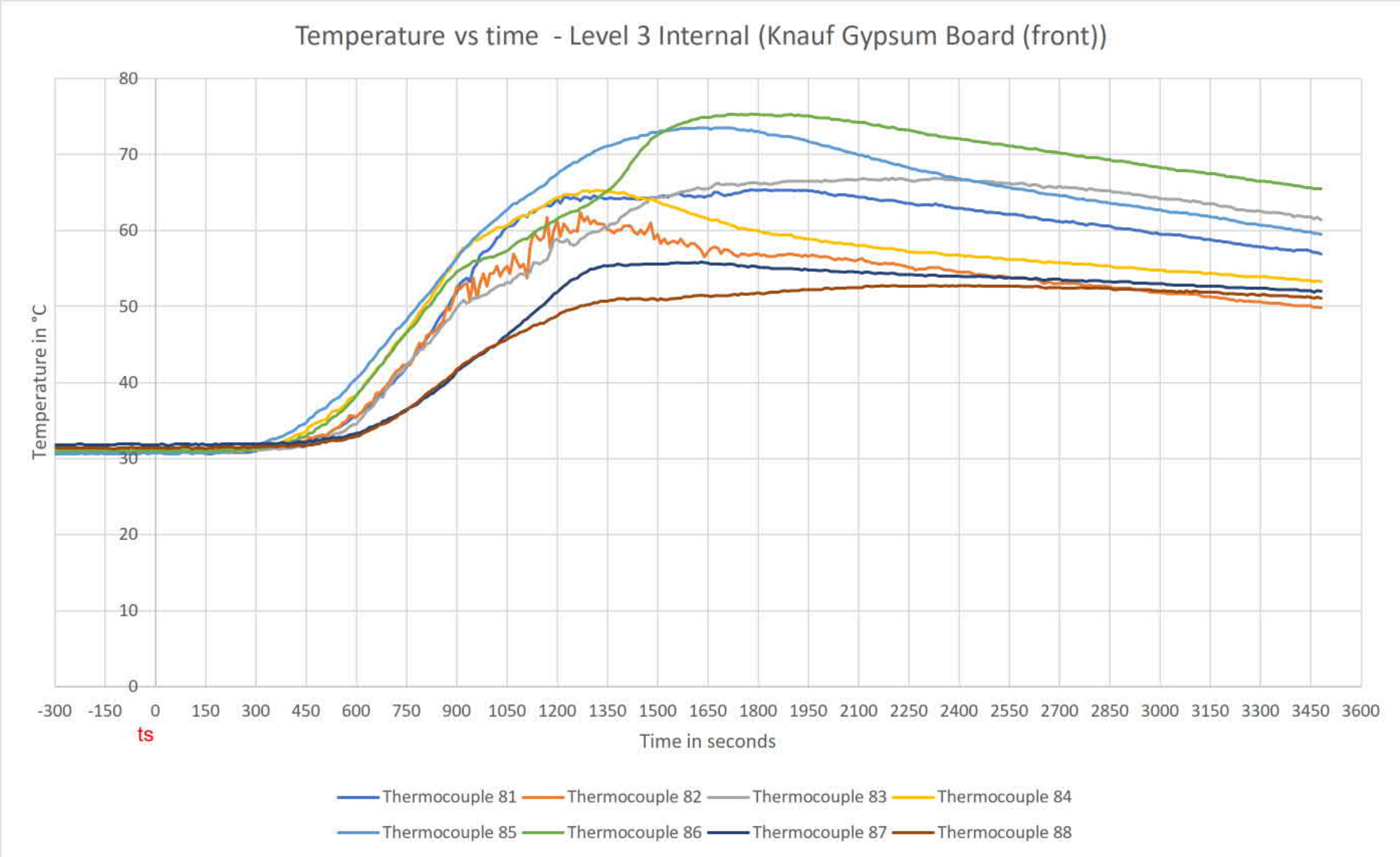


Figure 15 Level 3, internal, Knauf gypsum board (front) – temperature vs time

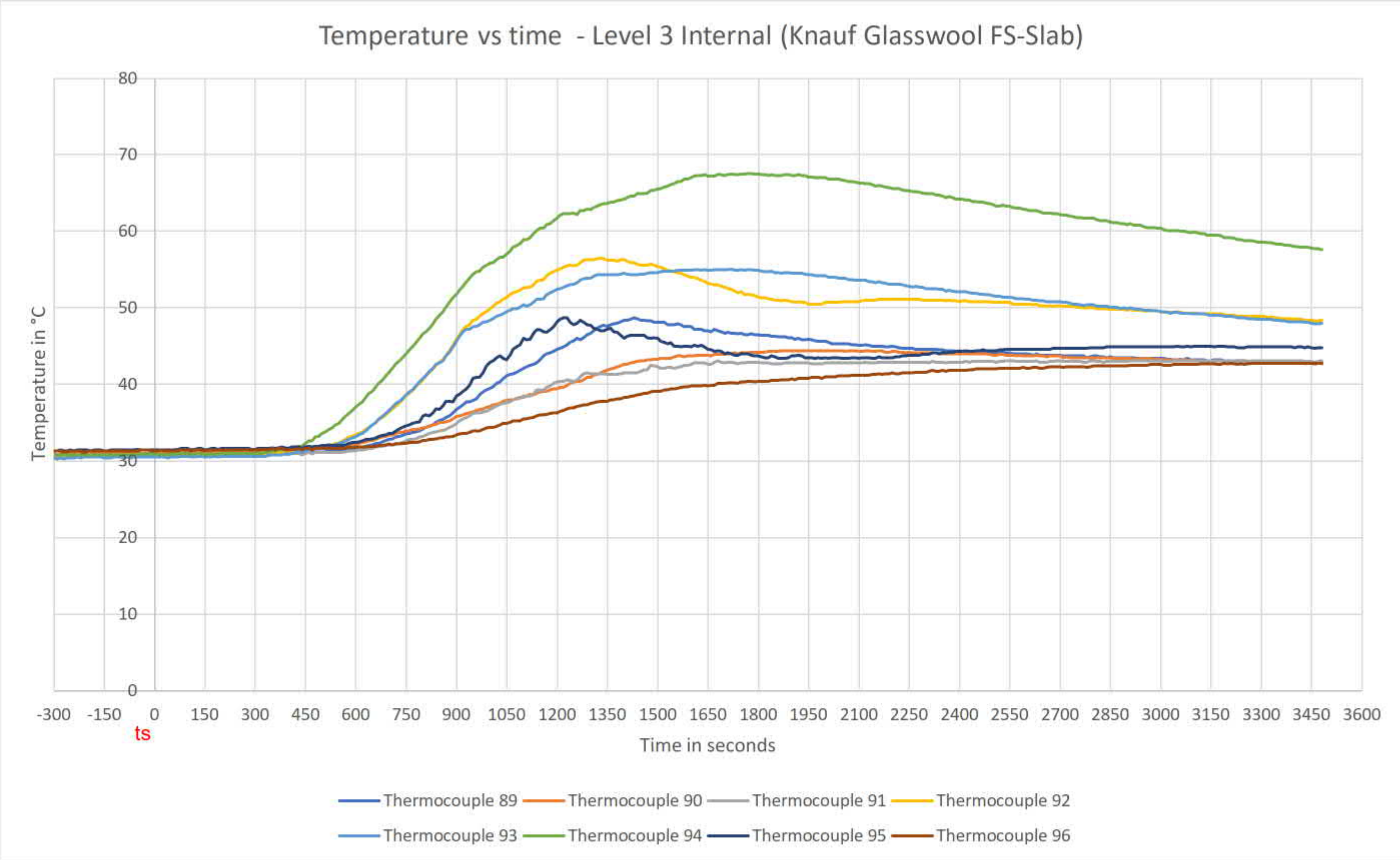


Figure 16 Level 3, internal, Knauf glasswool FS-Slab – temperature vs time

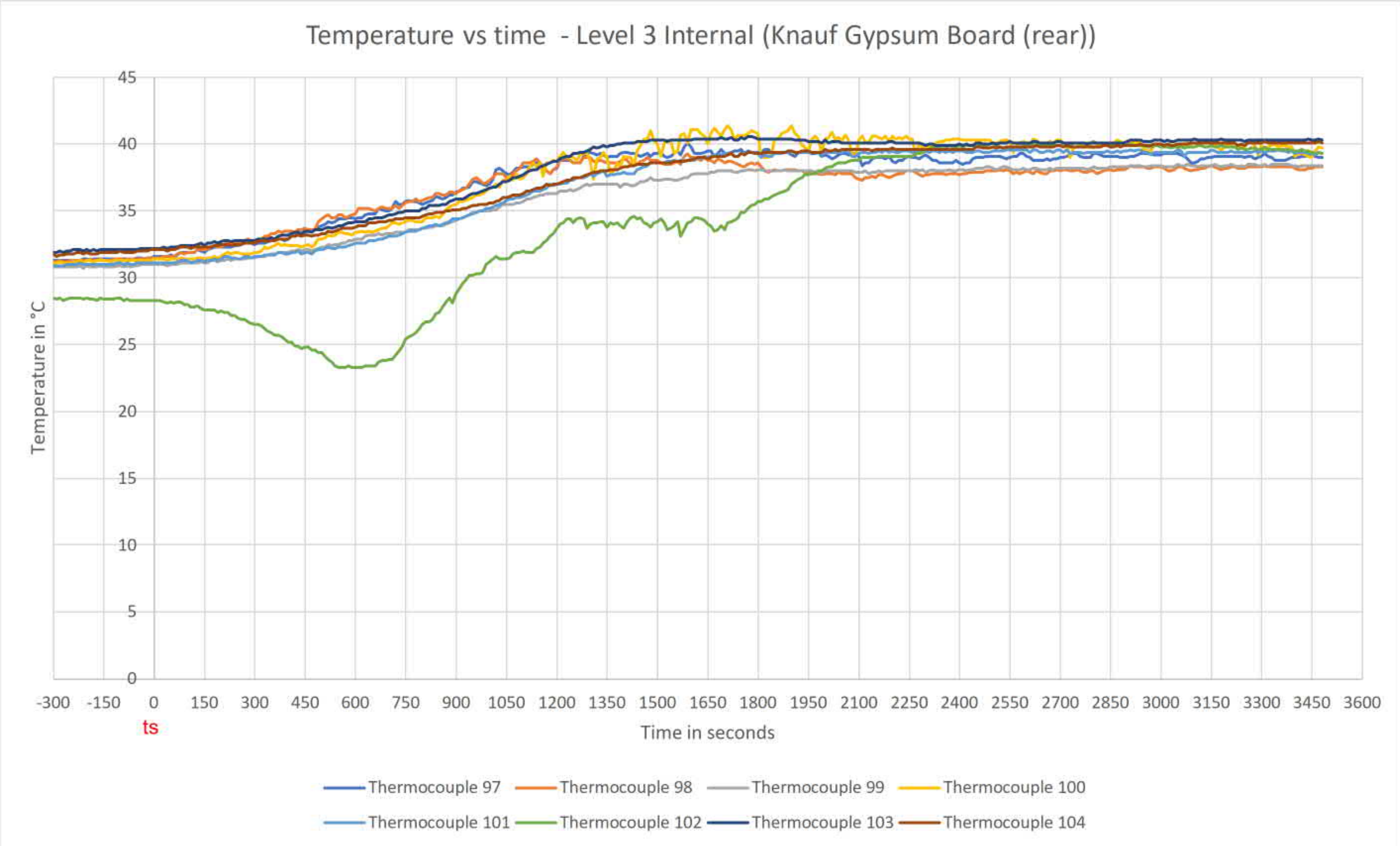


Figure 17 Level 3, internal, Knauf gypsum board (rear) – temperature vs time

Appendix C Photographs

C.1 Installation

Photo 2 Knauf SFS

Photo 3 Knauf glasswool FS-Slab

Photo 4 Tyvek membrane, Alubond RFS brackets, and Siderise cavity barriers

Photo 6 **Alubond FR-A2 panel**

Photo 7 Tyvek membrane

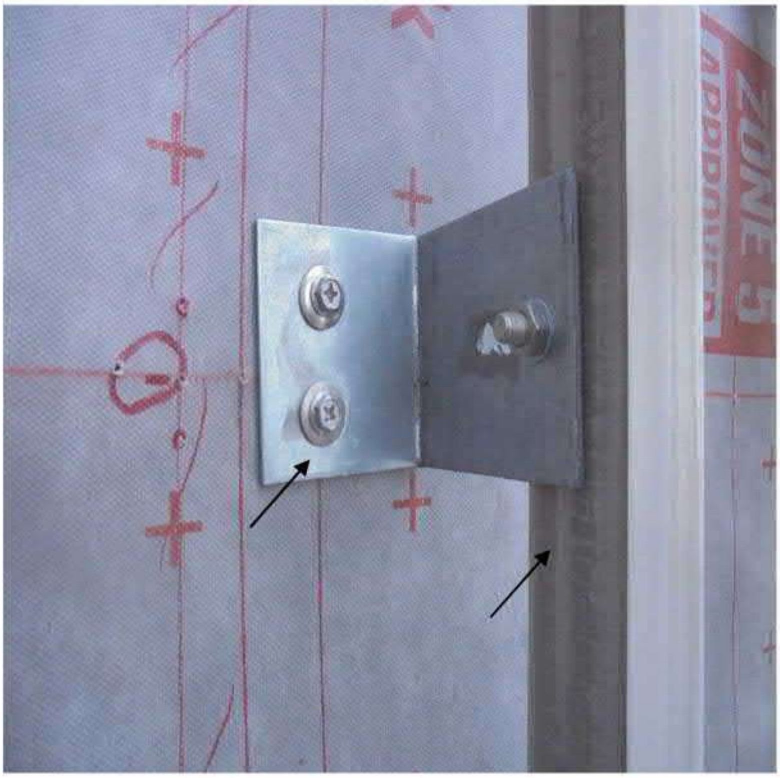


Photo 8 Alubond RFS wall bracket and Alubond RFS aluminium runner

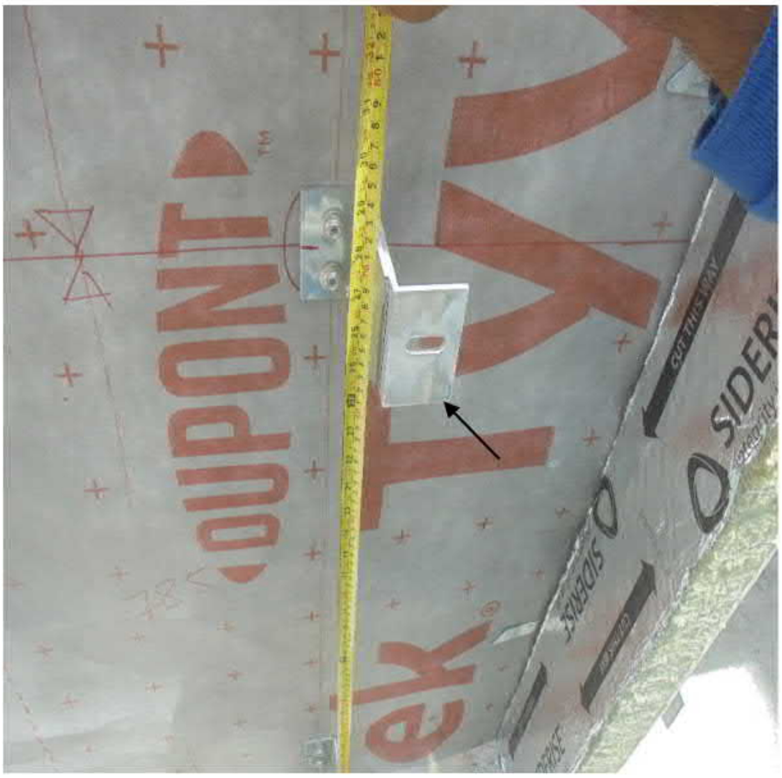


Photo 9 Alubond RFS aluminium horizontal support bracket fixed to the GI wall bracket

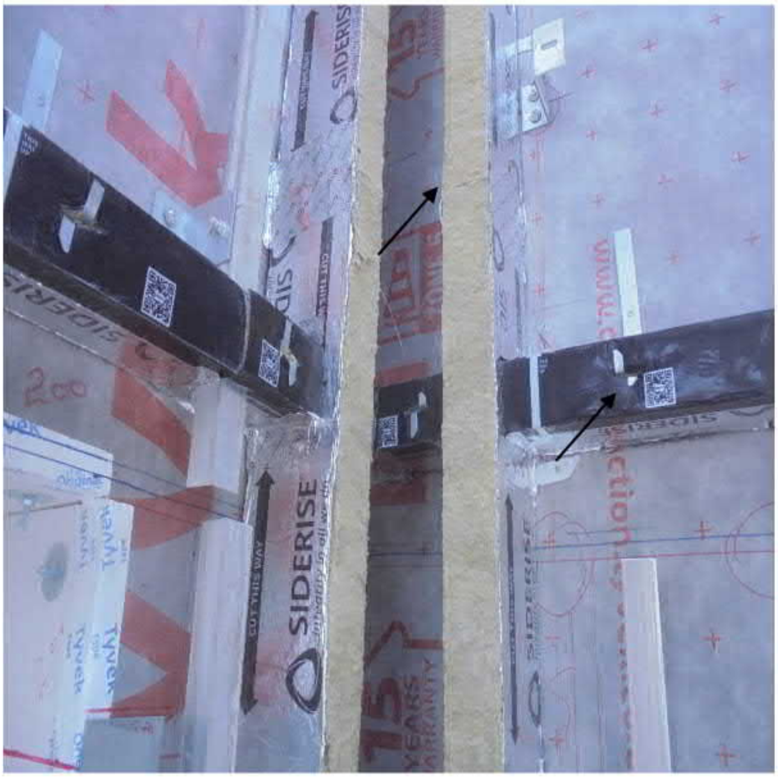


Photo 10 Siderise horizontal and vertical cavity barriers

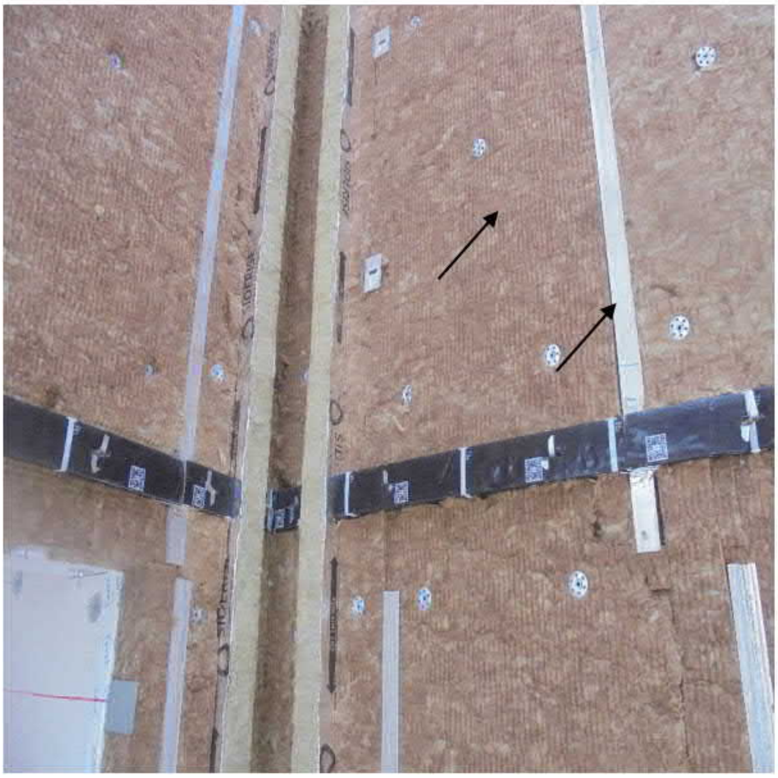


Photo 11 Knauf Rocksilk rainscreen slab insulation and Alubond RFS aluminium vertical runner

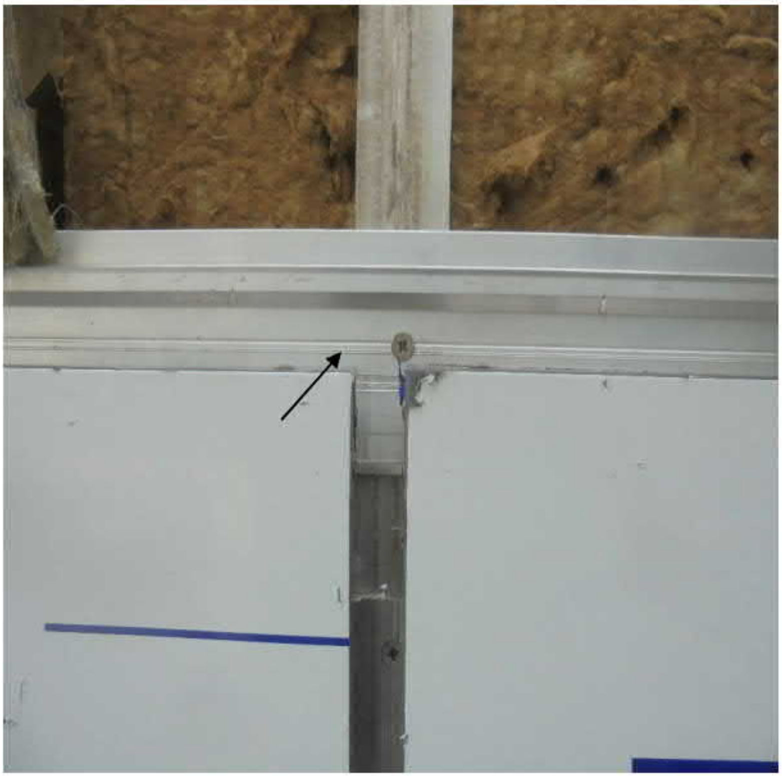


Photo 12 Alubond RFS aluminium panel attachment clip

Photo 13 **Alubond RFS aluminium flat bar**

Photo 14 **Alubond FR-A2 ACP fixing**

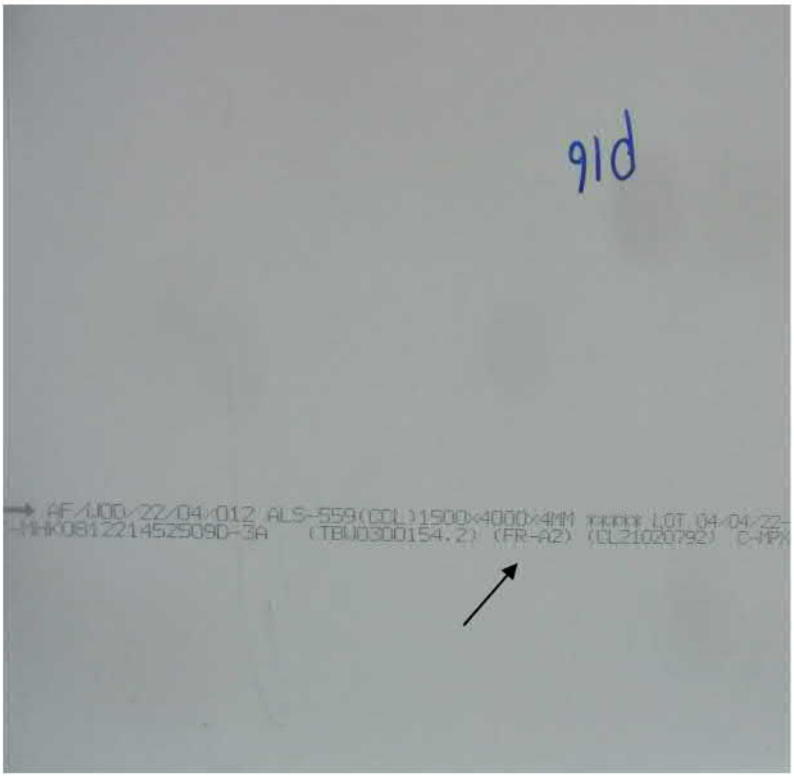


Photo 15 Panel identification mark on the interior surface of the cladding panel

Photo 16 Alubond RFS aluminium panel perimeter strip

Photo 17 Galvanized steel flashing around the combustion chamber

C.2 Testing

Photo 18 Coating of mid-panels up to 1 m above the combustion chamber started to peel off

Photo 19 Coating of panels up to 2 m above the combustion chamber started to peel off

Photo 20 Coating of mid-panels at 3 m above the combustion chamber started to peel off

Photo 21 Mid-panels at 1 m above the combustion chamber were partially consumed and insulation was exposed

Photo 22 **Mid-panels between 1 to 2 m above the combustion chamber were partially consumed and insulation exposed**

Photo 23 **Coating of mid-panels at 5 m above the combustion chamber started to peel off**

Photo 24 Flaming debris from mid panel at 1 m above the combustion chamber fell off

Photo 25 Debris of core from mid panels between 1 to 2 m above the combustion chamber fell off

Photo 26 Self-sustained flames on mid-panel at level 1

Photo 27 Debris of mid-panel at 3 m above the combustion chamber fell off

Photo 28 **Mid-panels at 3 m above the combustion chamber were partially consumed and insulation exposed**

C.3 Post-test

Photo 29 Alubond FR-A2 ACP cladding on completion of the test

Photo 30 **Siderise cavity barrier, Knauf Rocksilk insulation, Alubond RFS panel attachment clip and runners**

Photo 31 Dupont Tyvek Supro membrane and Alubond RFS brackets

Photo 32 **Knauf Type X gypsum board (front)**

Photo 33 Knauf glasswool FS-Slab

Photo 34 Siderise cavity barriers, Knauf Rocksilk insulation and Alubond RFS railings at level 1

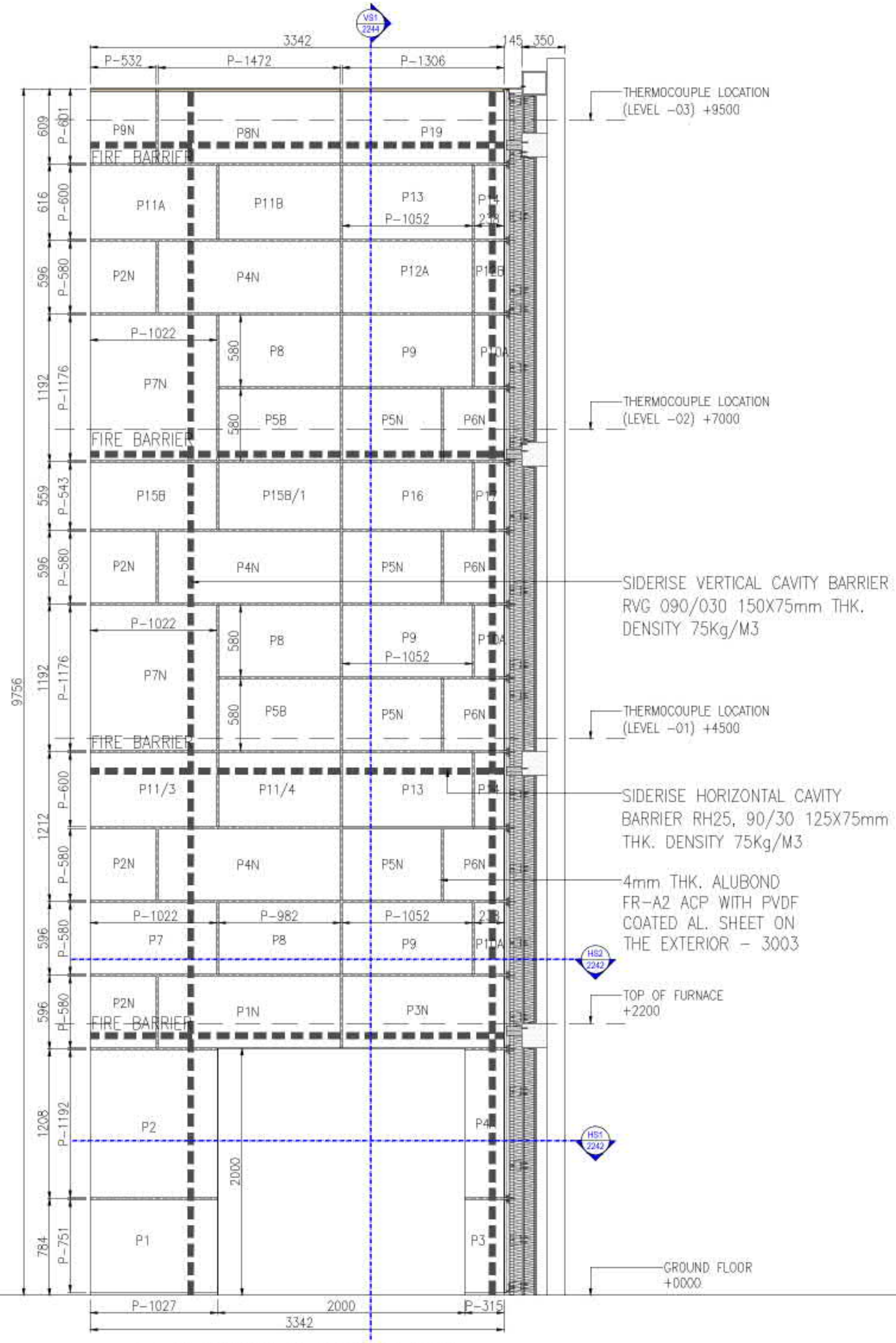
Photo 35 Siderise cavity barriers, Knauf Rocksilk insulation and Alubond RFS railings at level 2

Photo 36 Siderise cavity barriers and Alubond RFS railings at level 3

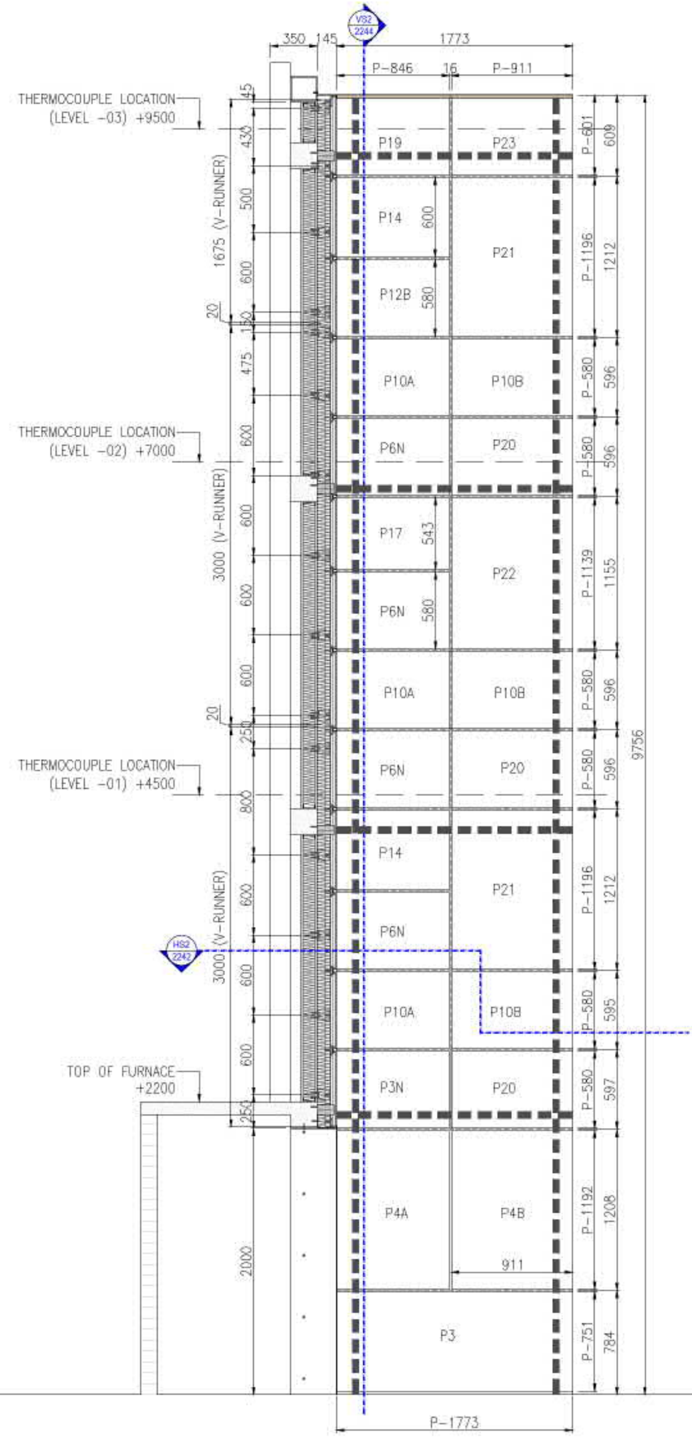
Appendix D Drawings

The following nine unpaginated pages include copies of Eurocon Building Industries FZE drawings numbered:

- BS-8414-P2-2241 Rev. 03
- BS-8414-P2-2242 Rev. 03
- BS-8414-P2-2243 Rev. 03
- BS-8414-P2-2244 Rev. 03
- BS-8414-P2-2245 Rev. 03
- BS-8414-P2-2246 Rev. 03
- BS-8414-P2-2247 Rev. 03
- BS-8414-P2-2248 Rev. 03
- BS-8414-P2-2249 Rev. 03



02 / 2241 **MAIN WALL ELEVATION**
SCALE 1:20



01 / 2241 **RETURN WALL ELEVATION**
SCALE 1:20

Al-Futtaim element
Al Futtaim Element Materials Technology Dubai L.L.C
Facade Testing & Advisory Services Division

CODE	DESCRIPTION
B1	90X50X75X4mm THK. G.I BRACKET
B3	90X50X75X4mm THK. G.I BRACKET
B4	100X100X75X4mm THK. G.I BRACKET
ST	STUD DISTANCE
P	ACP PANEL DIMENSION

Purpose of Issue	Rev.	Date	Approved
For Approval	00	06-06-2022	
REVISED AS PER COMMENTS	01	20-06-2022	
REVISED AS PER COMMENTS	02	25-06-2022	
ADD MATERIAL CODES	03	04-08-2022	

SYSTEM PROVIDER:
Alubond U.S.A Fire Rated Metal Composites
ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)

ACP SUPPLIER:
Alubond U.S.A Fire Rated Metal Composites

FABRICATOR & INSTALLATIONS
Eurocon Building Industries L.L.C
AL JURF INDUSTRIAL - 1
AJMAN - U.A.E

FIRE STOP SUPPLIER:
SIDERISE
SIDERISE, Forge Industrial Estate, Maesteg, UK. CF34 0AY
T: +44 (0)1656 730833

INSULATION SUPPLIER:
KNAUF INSULATION

JOB TITLE:
BS-8414 PART-2
SYSTEM DEVELOPMENT -
ARFS ALUBOND RAIN SCREEN
FACADE SYSTEM

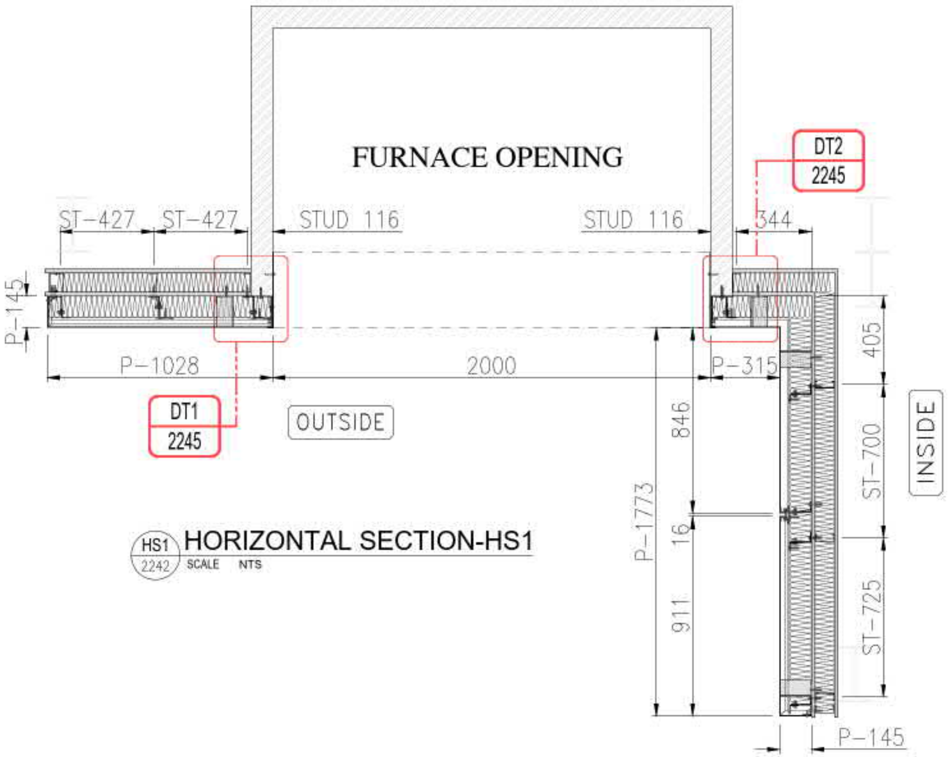
DRG. TITLE:
BS-8414 PART-2 SYSTEM
DEVELOPMENT- ARFS
ACP PANEL ELEVATIONS

FIRE LAB:
Al-Futtaim element

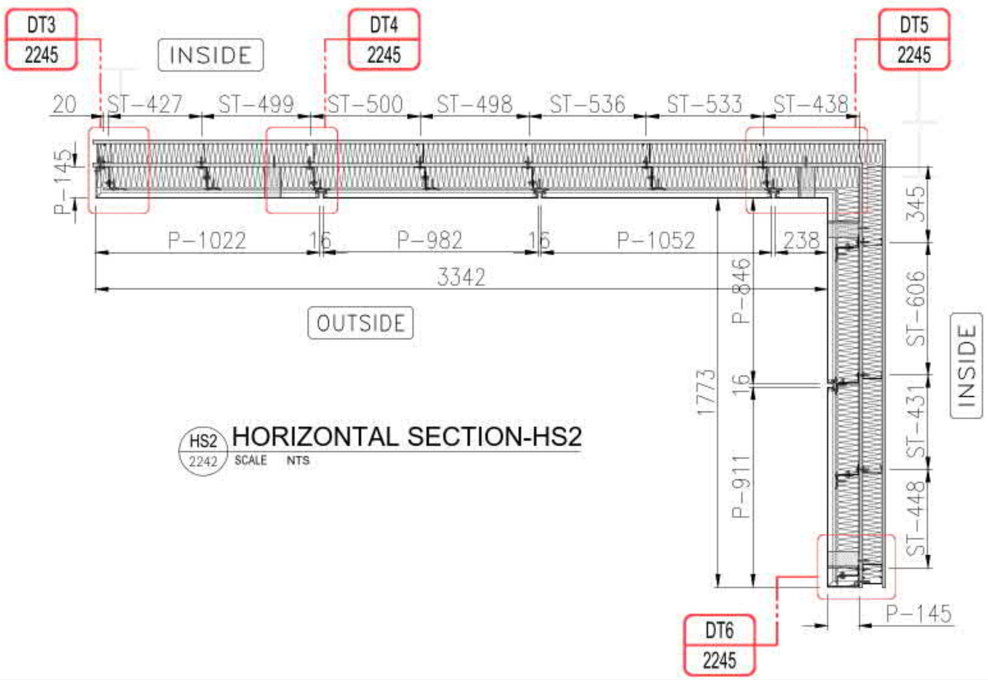
Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

Drawing Number: BS-8414-P2-2241
Rev: 03

CODE	DESCRIPTION
B1	90X50X75X4mm THK. G.I BRACKET
B3	90X50X75X4mm THK. G.I BRACKET
B4	100X100X75X4mm THK. G.I BRACKET
ST	STUD DISTANCE
P	ACP PANEL DIMENSION



HS1 2242 HORIZONTAL SECTION-HS1
SCALE NTS



HS2 2242 HORIZONTAL SECTION-HS2
SCALE NTS

Purpose of Issue	Rev.	Date	Approved
For Approval	00	06-06-2022	
REVISED AS PER COMMENTS	01	20-06-2022	
REVISED AS PER COMMENTS	02	25-06-2022	
ADD MATERIAL CODES	03	04-08-2022	

SYSTEM PROVIDER:
Alubond U.S.A Fire Rated Metal Composites
ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)

ACP SUPPLIER:
Alubond U.S.A Fire Rated Metal Composites

FABRICATOR & INSTALLATIONS
Eurocon Building Industries L.L.C
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AJMAN - U.A.E

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INSULATION SUPPLIER:
KNAUF INSULATION

JOB TITLE:
BS-8414 PART-2
SYSTEM DEVELOPMENT - ARFS
ALUBOND RAIN SCREEN
FACADE SYSTEM

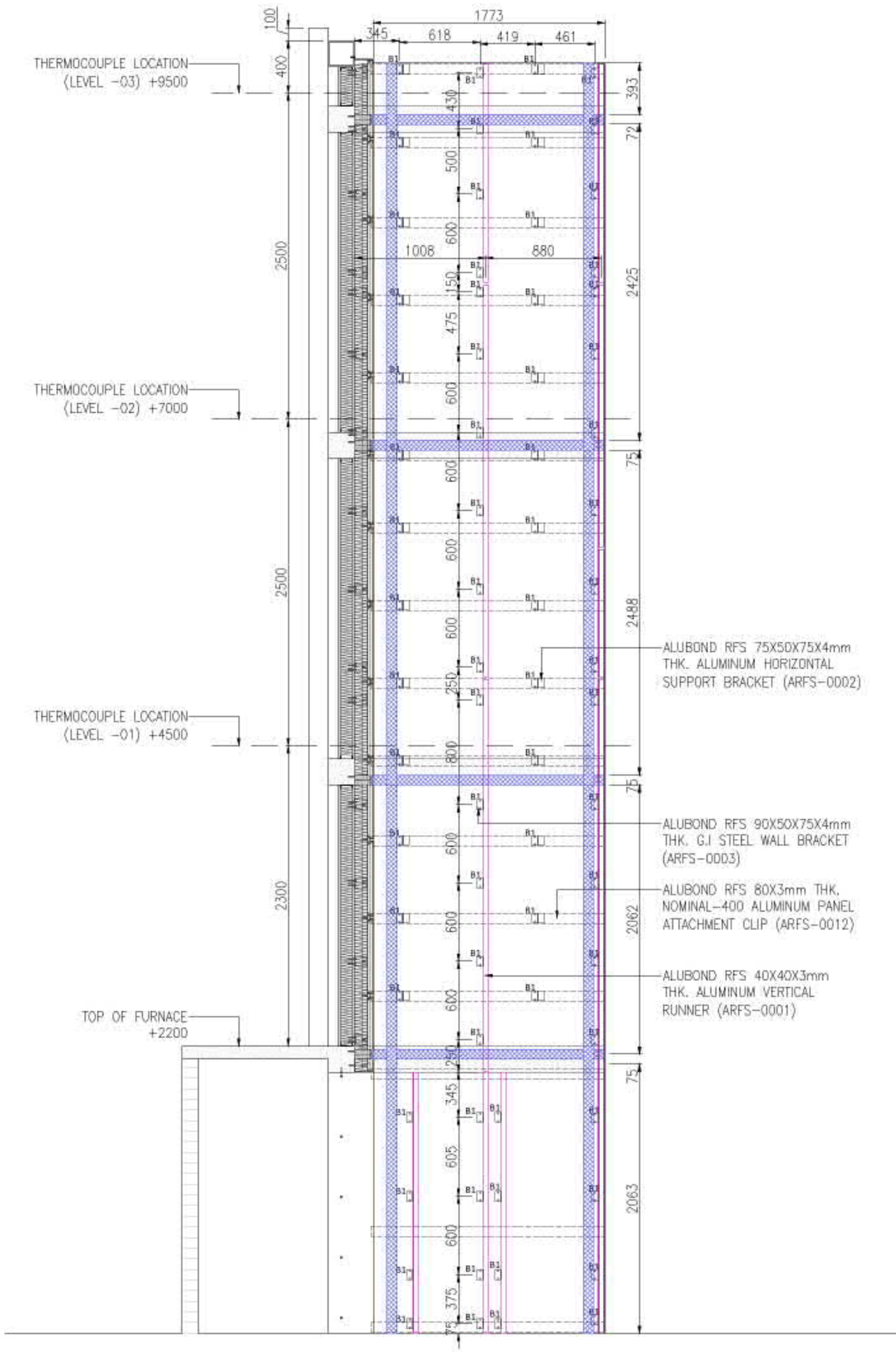
DRG. TITLE:
BS-8414 PART-2 SYSTEM
DEVELOPMENT- ARFS
HORIZONTAL SECTION HS1 & HS2

FIRE LAB.:
الفطيم Al-Futtaim اليمنت element

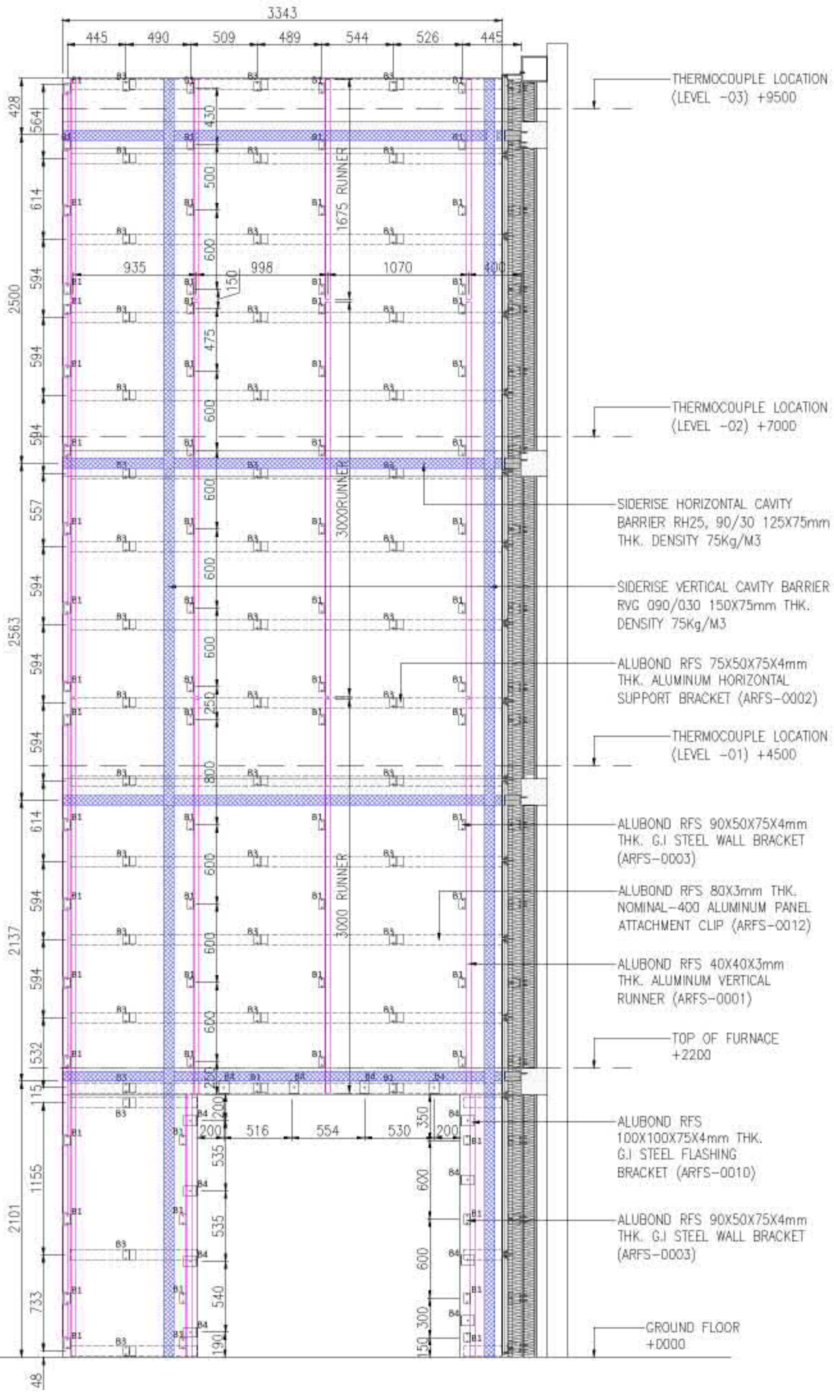
الفطيم Al-Futtaim اليمنت element
Al Futtaim Element Materials Technology Dubai L.L.C
Facade Testing & Advisory Services Division

Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

Drawing Number: BS-8414-P2-2242
Rev: 03



01 2243 **RETURN WALL RUNNER ELEVATION**
SCALE 1:20



02 2243 **MAIN WALL RUNNER ELEVATION**
SCALE 1:20

CODE	DESCRIPTION
B1	90X50X75X4mm THK. G.I BRACKET
B3	90X50X75X4mm THK. G.I BRACKET
B4	100X100X75X4mm THK. G.I BRACKET
ST	STUD DISTANCE
P	ACP PANEL DIMENSION

Purpose of Issue	Rev.	Date	Approved
For Approval	00	06-06-2022	
REVISED AS PER COMMENTS	01	20-06-2022	
REVISED AS PER COMMENTS	02	25-06-2022	
ADD MATERIAL CODES	03	04-08-2022	

SYSTEM PROVIDER:
Alubond U.S.A Fire Rated Metal Composites
ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)

ACP SUPPLIER:
Alubond U.S.A Fire Rated Metal Composites

FABRICATOR & INSTALLATIONS
Eurocon Building Industries L.L.C
AL JURF INDUSTRIAL - 1
AJMAN - U.A.E

FIRE STOP SUPPLIER:
SIDERISE, Forge Industrial Estate, Maesteg, UK, CF34 0AY
T: +44 (0)1656 730833

INSULATION SUPPLIER:
KNAUF INSULATION

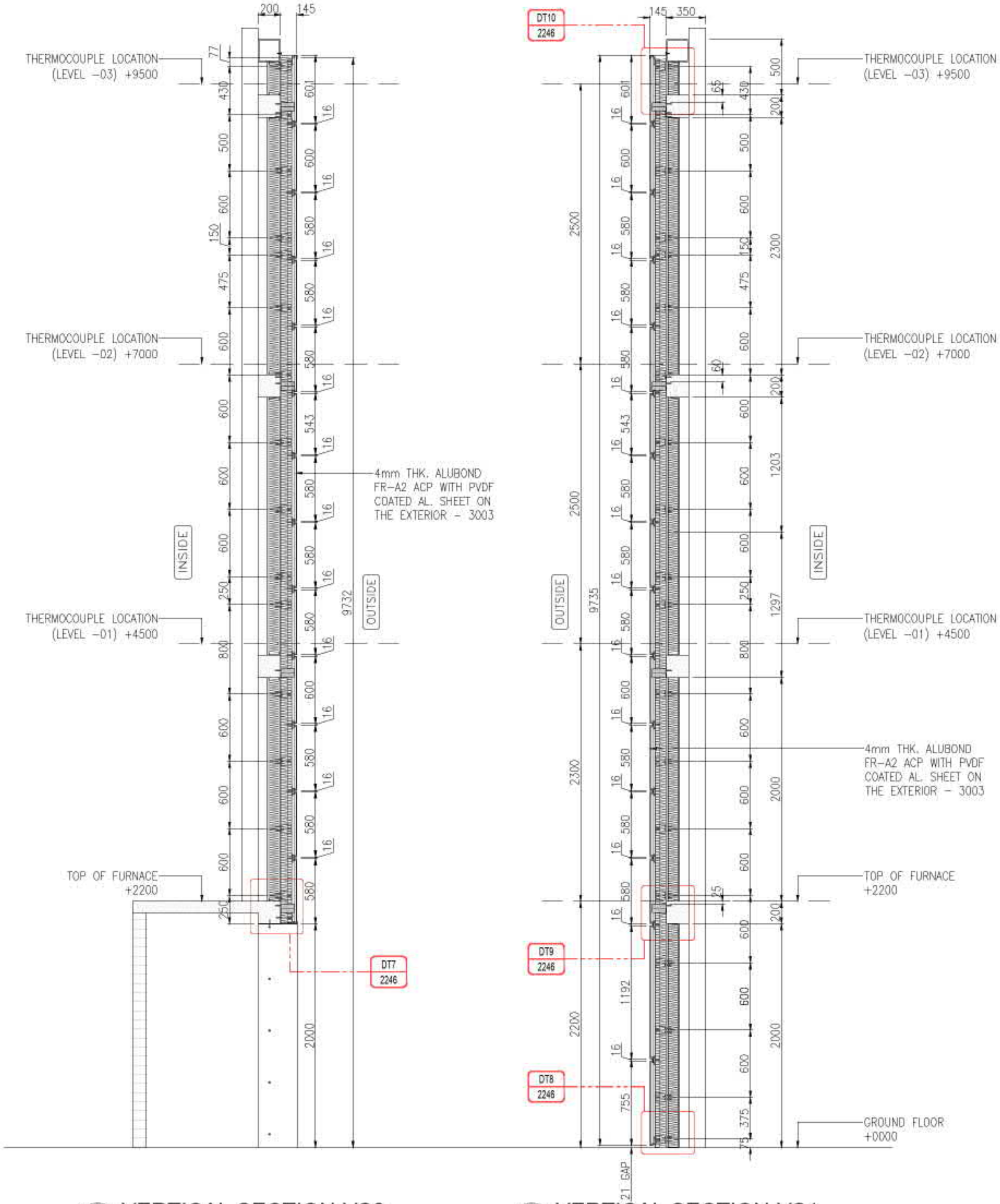
JOB TITLE:
BS-8414 PART-2
SYSTEM DEVELOPMENT - ARFS
ALUBOND RAIN SCREEN FACADE SYSTEM

DRG. TITLE:
BS-8414 PART-2 SYSTEM DEVELOPMENT- ARFS
ALUMINIUM RUNNER & FIRE BARRIER ELEVATION

FIRE LAB.:
الفطيم Al-Futtaim اليمنت element

Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

Drawing Number: BS-8414-P2-2243
Rev: 03



VS2 2244 VERTICAL SECTION-VS2
SCALE NTS

VS1 2244 VERTICAL SECTION-VS1
SCALE NTS

CODE	DESCRIPTION
B1	90X50X75X4mm THK. G.I BRACKET
B3	90X50X75X4mm THK. G.I BRACKET
B4	100X100X75X4mm THK. G.I BRACKET
ST	STUD DISTANCE
P	ACP PANEL DIMENSION

Purpose of Issue	Rev.	Date	Approved
For Approval	00	06-06-2022	
REVISED AS PER COMMENTS	01	20-06-2022	
REVISED AS PER COMMENTS	02	25-06-2022	
ADD MATERIAL CODES	03	04-08-2022	

SYSTEM PROVIDER:
Alubond U.S.A Fire Rated Metal Composites
ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)

ACP SUPPLIER:
Alubond U.S.A Fire Rated Metal Composites

FABRICATOR & INSTALLATIONS
Eurocon Building Industries L.L.C
AL JURF INDUSTRIAL - 1
AJMAN - U.A.E

FIRE STOP SUPPLIER:
SIDERISE. Forge Industrial Estate, Maesteg, UK, CF34 0AY
T: +44 (0)1656 730833

INSULATION SUPPLIER:
KNAUF INSULATION

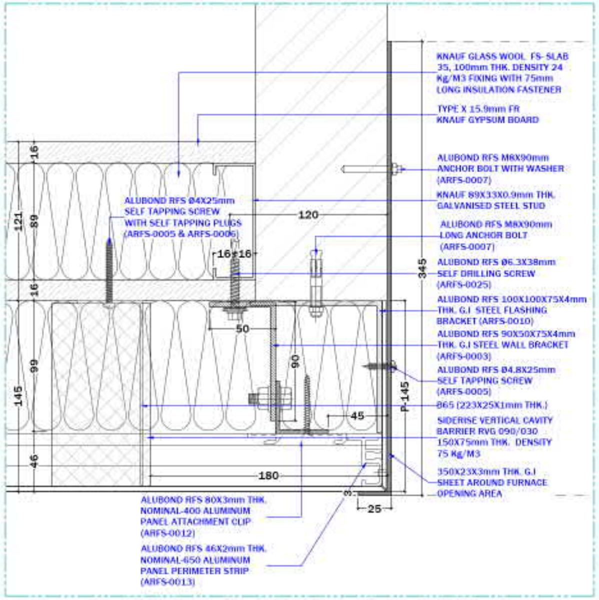
JOB TITLE:
BS-8414 PART-2
SYSTEM DEVELOPMENT - ARFS
ALUBOND RAIN SCREEN FACADE SYSTEM

DRG. TITLE:
BS-8414 PART-2 SYSTEM DEVELOPMENT- ARFS
VERTICAL SECTION VS1 & VS2

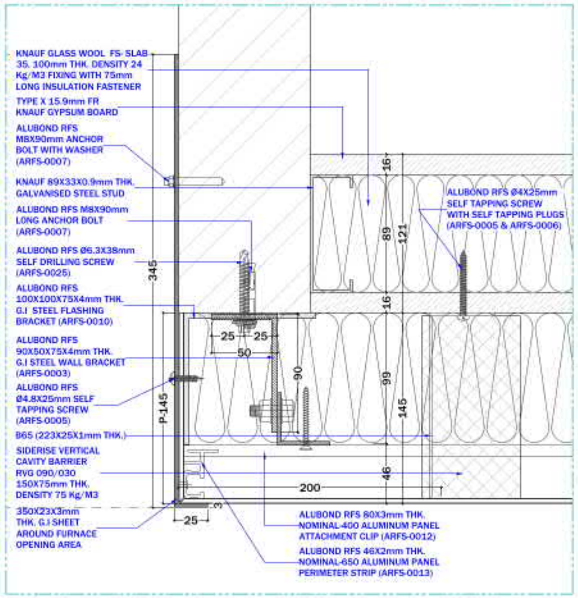
FIRE LAB:
الفطيم Al-Futtaim اليمنت element

Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

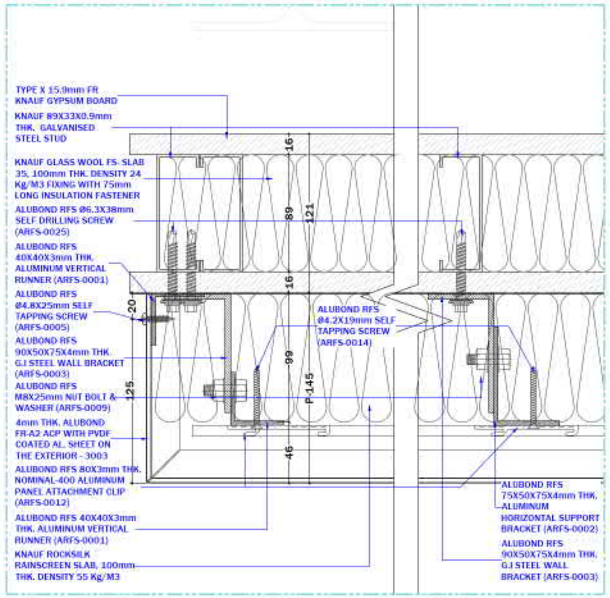
Drawing Number	Rev
BS-8414-P2-2244	03



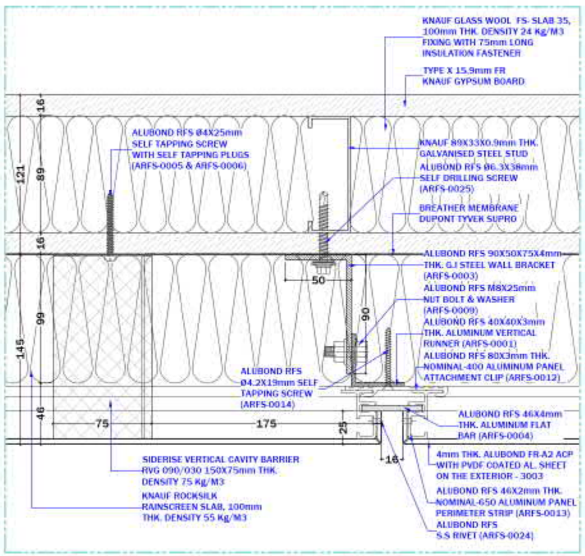
DT-1 2245 DETAIL - D1 SCALE 1:2



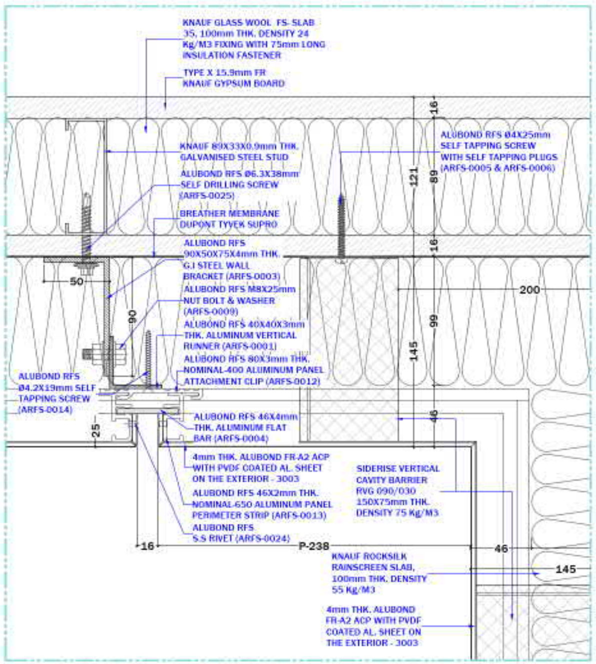
DT-2 2245 DETAIL - D2 SCALE 1:2



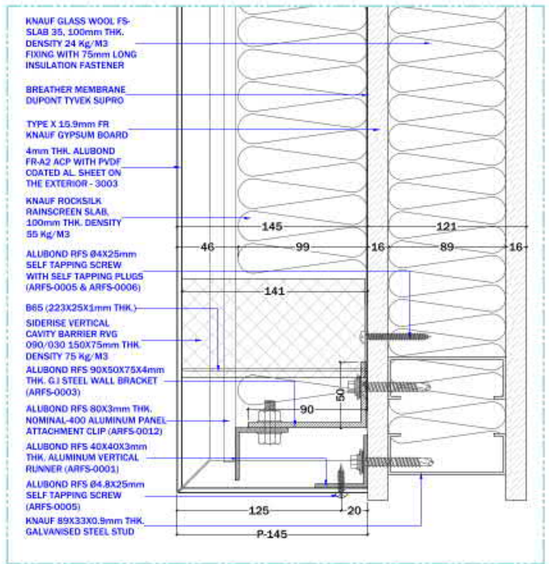
DT-3 2245 DETAIL - D3 SCALE 1:2



DT-4 2245 DETAIL - D4 SCALE 1:2



DT-5 2245 DETAIL - D5 SCALE 1:2



DT-6 2245 DETAIL - D6 SCALE 1:2

Purpose of Issue	Rev.	Date	Approved
For Approval	00	06-06-2022	
REVISED AS PER COMMENTS	01	20-06-2022	
REVISED AS PER COMMENTS	02	25-06-2022	
ADD MATERIAL CODES	03	04-08-2022	

SYSTEM PROVIDER:

Alubond U.S.A Fire Rated Metal Composites

ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)

ACP SUPPLIER:

Alubond U.S.A Fire Rated Metal Composites

FABRICATOR & INSTALLATIONS

Eurocon Building Industries L.L.C
AL JURF INDUSTRIAL - 1
AJMAN - U.A.E

FIRE STOP SUPPLIER:

SIDERISE
SIDERISE, Forge Industrial Estate, Maesteg, UK, CF34 0AY
T: +44 (0)1656 730833

INSULATION SUPPLIER:

KNAUF INSULATION

JOB TITLE:

BS-8414 PART-2
SYSTEM DEVELOPMENT - ARFS
ALUBOND RAIN SCREEN
FACADE SYSTEM

DRG. TITLE:

BS-8414 PART-2 SYSTEM
DEVELOPMENT- ARFS
ENLARGE DETAILS (DT-1 TO DT-6)

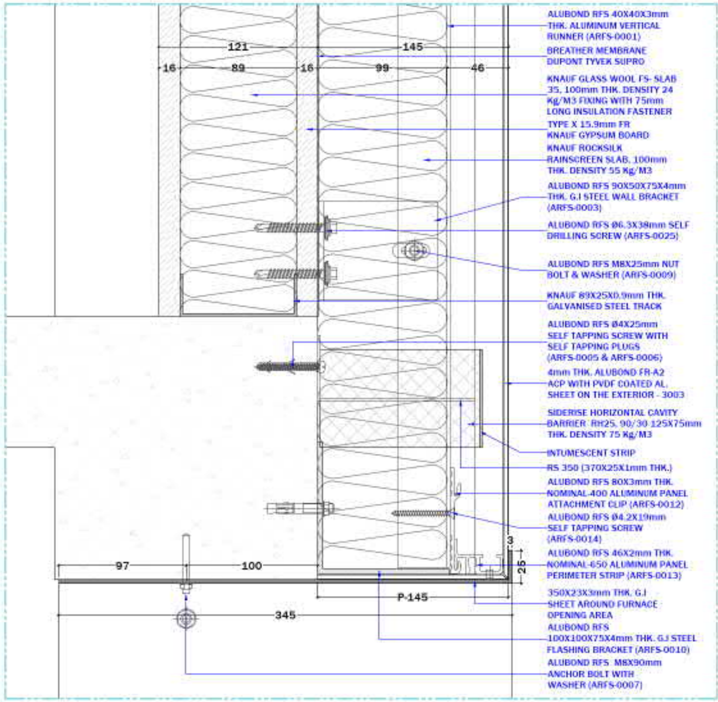
FIRE LAB.:

الفطيم Al-Futtaim اليمنت element

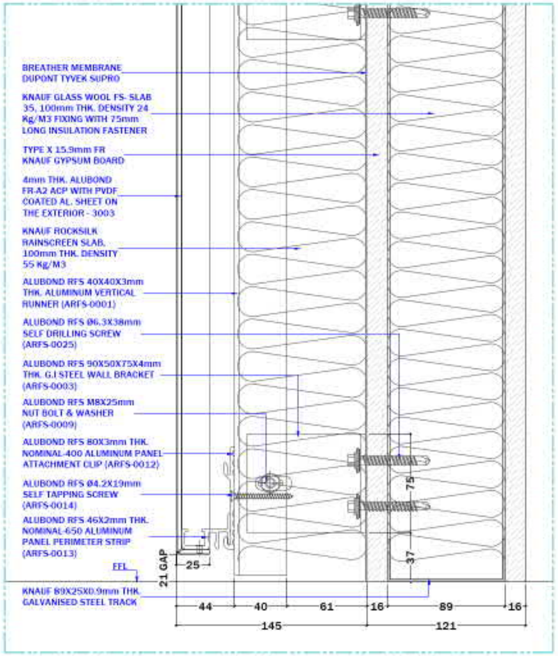
Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

Drawing Number: BS-8414-P2-2245

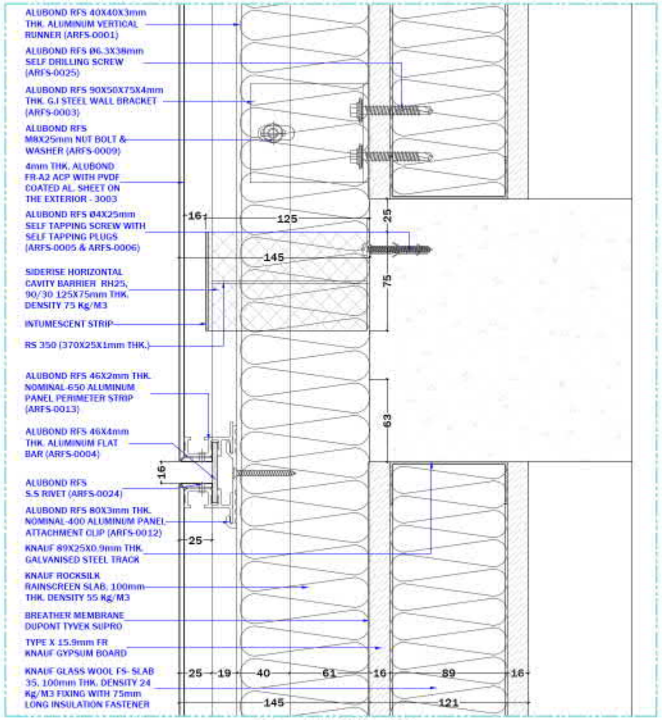
Rev: 03



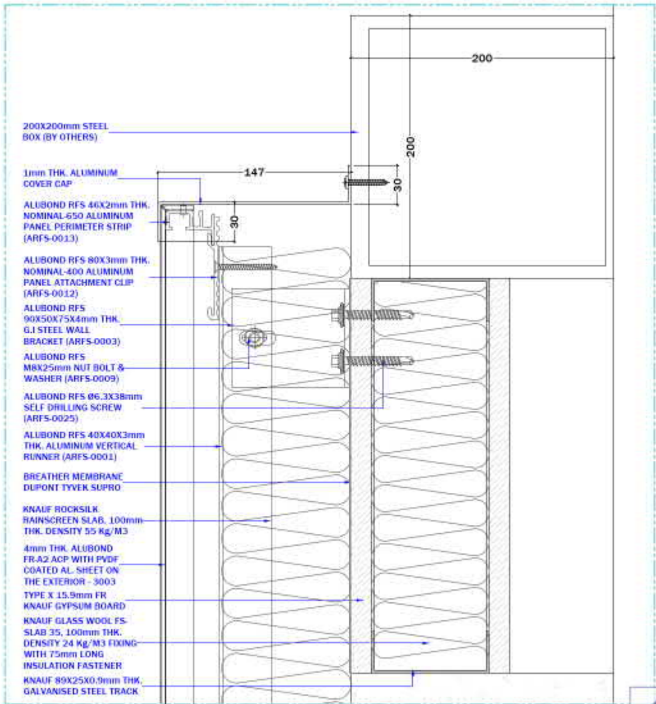
DT-7 2246 DETAIL - D7 SCALE 1:2



DT-8 2246 DETAIL - D8 SCALE 1:2



DT-9 2246 DETAIL - D9 SCALE 1:2



DT-10 2246 DETAIL - D10 SCALE 1:2

Al-Futtaim Element Materials Technology Dubai L.L.C
Facade Testing & Advisory Services Division

Purpose of Issue	Rev.	Date	Approved
For Approval	00	06-06-2022	
REVISED AS PER COMMENTS	01	20-06-2022	
REVISED AS PER COMMENTS	02	25-06-2022	
ADD MATERIAL CODES	03	04-08-2022	

SYSTEM PROVIDER:
Alubond U.S.A Fire Rated Metal Composites
ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)

ACP SUPPLIER:
Alubond U.S.A Fire Rated Metal Composites

FABRICATOR & INSTALLATIONS
Eurocon Building Industries L.L.C
AL JURF INDUSTRIAL - 1
AJMAN - U.A.E

FIRE STOP SUPPLIER:
SIDERISE
SIDERISE, Forge Industrial Estate, Maesteg, UK, CF34 0AY
T: +44 (0)1656 730833

INSULATION SUPPLIER:
KNAUF INSULATION

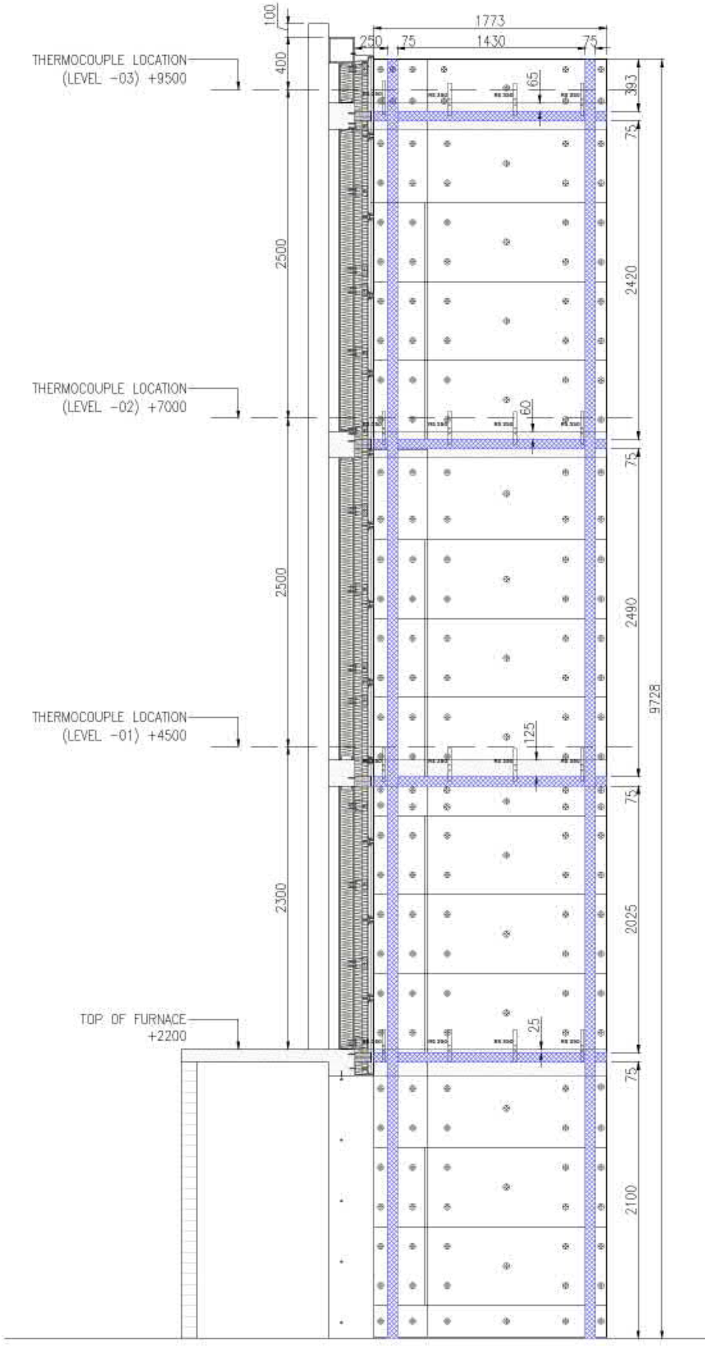
JOB TITLE:
BS-8414 PART-2
SYSTEM DEVELOPMENT - ARFS
ALUBOND RAIN SCREEN
FACADE SYSTEM

DRG. TITLE:
BS-8414 PART-2 SYSTEM
DEVELOPMENT- ARFS
ENLARGE DETAILS (DT-7 TO DT-10)

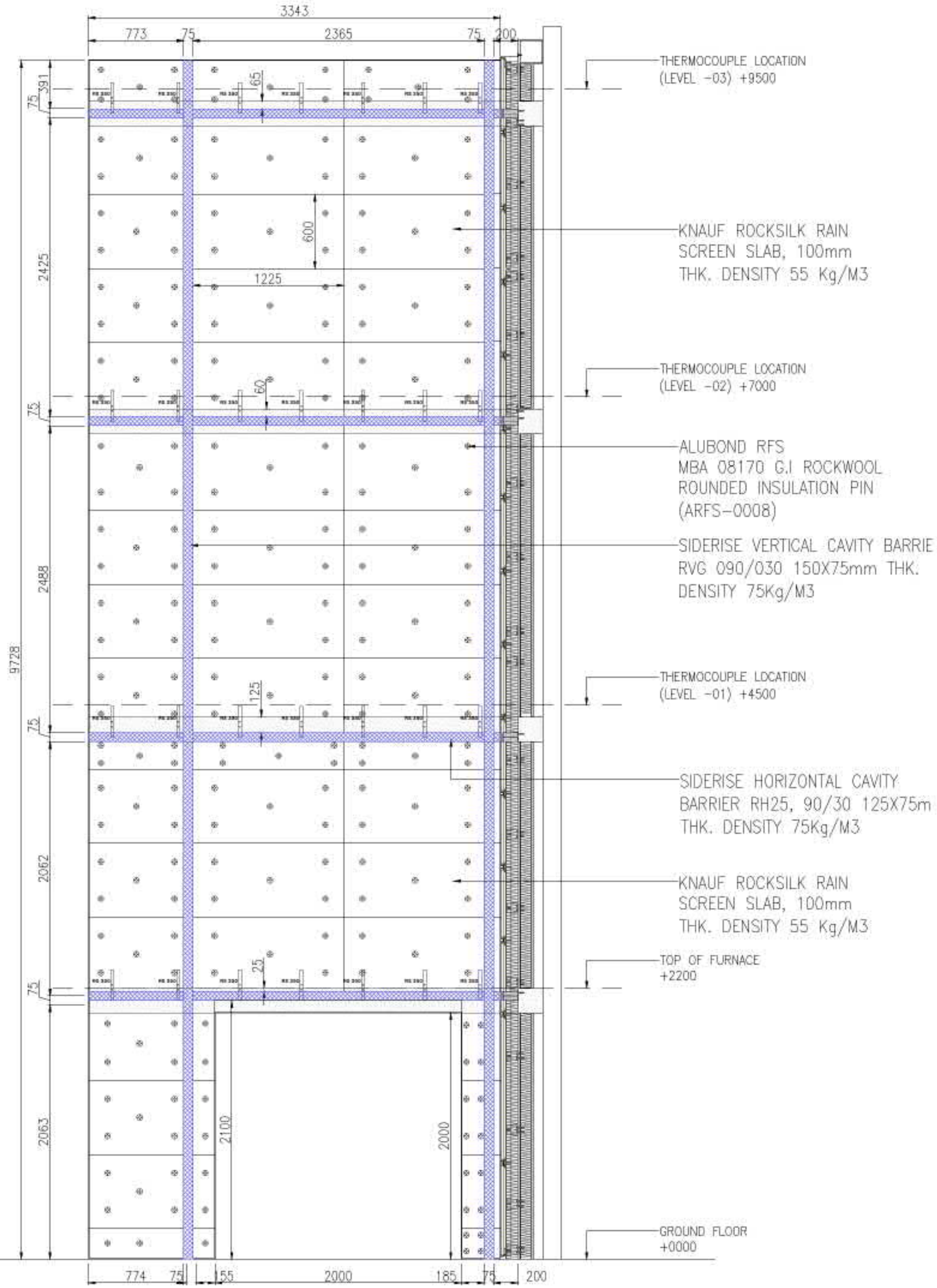
FIRE LAB.:
الفطيم Al-Futtaim اليمنت element

Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

Drawing Number	Rev
BS-8414-P2-2246	03



01
2243
RETURN WALL INSULATION ELEVATION
SCALE 1:20



02
2243
MAIN WALL INSULATION ELEVATION
SCALE 1:20

CODE	DESCRIPTION
B1	90X50X75X4mm THK. G.I BRACKET
B3	90X50X75X4mm THK. G.I BRACKET
B4	100X100X75X4mm THK. G.I BRACKET
ST	STUD DISTANCE
P	ACP PANEL DIMENSION

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FACADE SYSTEM

DRG. TITLE:
BS-8414 PART-2 SYSTEM
DEVELOPMENT- ARFS
FIRE BARRIER & KNAUF INSULATION ELEVATION

FIRE LAB:
الفطيم Al-Futtaim اليمنت element

Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

Drawing Number: BS-8414-P2-2247
Rev: 03

Al-Futtaim element
Al Futtaim Element Materials Technology Dubai L.L.C
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ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)			
SL/NO	ALUBOND CODE	IMAGES	PRODUCT AND MATERIAL INFORMATION
1	FR-A2-3003		4mm THK. ALUBOND FR-A2 ACP WITH PVDF COATED AL. SHEET ON THE EXTERIOR - 3003
2	ARFS-0001		ALUBOND RFS ALUMINIUM VERTICAL RUNNER PROFILE (40X40X3mm) ALLOY 6063
3	ARFS-0002		ALUBOND RFS ALUMINIUM HORIZONTAL SUPPORT BRACKET (75X50X75X4mm)
4	ARFS-0003		ALUBOND RFS 90X50X75X4mm THK. G.I STEEL WALL BRACKET
5	ARFS-0004		ALUBOND RFS 46X4mm THK. ALUMINUM FLAT BAR
6	ARFS-0005 & ARFS-0006		ALUBOND RFS Ø4X25mm SELF TAPPING SCREW WITH SELF TAPPING PLUGS
7	ARFS-0007		ALUBOND RFS M8X90mm LONG ANCHOR BOLT
8	ARFS-0008		ALUBOND RFS MBA 08170 G.I ROCKWOOL ROUNDED INSULATION PIN
9	ARFS-0009		ALUBOND RFS M8X25mm NUT BOLT & WASHER
10	ARFS-0010		ALUBOND RFS 100X100X75X4mm THK. G.I STEEL FLASHING BRACKET
11	ARFS-0011		ALUBOND RFS ROCKWOOL INSTALLTION PIN SELF ADHESIVE (STICK PIN - 75mm)
12	ARFS-0012		ALUBOND RFS 80X3mm THK. NOMINAL-400 ALUMINUM PANEL ATTACHMENT CLIP
13	ARFS-0013		ALUBOND RFS 46X2mm THK. NOMINAL-650 ALUMINUM PANEL PERIMETER STRIP
14	ARFS-0014		ALUBOND RFS Ø4.2X19mm SELF TAPPING SCREW

ALUBOND RAIN SCREEN FACADE SYSTEM (ARFS)			
SL/NO	ALUBOND CODE	IMAGES	PRODUCT AND MATERIAL INFORMATION
15	ARFS-0022		ALUBOND RFS CARTRIDGE NAILS, WASHER & GRIPPER
16	ARFS-0024		ALUBOND RFS 4X14mm S.S RIVET
17	ARFS-0025		ALUBOND RFS Ø6.3X38mm SELF DRILLING SCREW
18	-		BREATHER MEMBRANE DUPONT TYVEK SUPRO
19	-		SIDERISE CAVITY BARRIER HORIZONTAL RH25G-90/30 125X75mm THK. DENSITY 75 Kg/M3, INTUMESCENT STRIP
20	-		SIDERISE CAVITY BARRIER VERTICAL RV-090/030 150X75mm THK. DENSITY 75 Kg/M3
21	-		KNAUF ROCKSILK RAINSCREEN SLAB, 100mm THK. DENSITY 55 Kg/M3
22	-		KNAUF GLASS WOOL FS- SLAB 35, 100mm THK. DENSITY 24 Kg/M3, FIXING WITH 75mm LONG INSULATION FASTENER
23	-		RS350 (370X25X1mmTHK.)
24	-		B65 (223X25X1mm THK.)
25	-		KNAUF 89X33X0.9mm THK. GALVANISED STEEL STUD KNAUF 89X25X0.9mm THK. GALVANISED STEEL TRACK
26	-		TYPE X 15.9mm FR KNAUF GYPSUM BOARD
27	-		ALUMINIUM ENCLOSER 30X147X30X1mm THK.
28	-		350X23X3mm THK. G.I SHEET AROUND FURNACE OPENING AREA (6000X370mm)

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ACP SUPPLIER:
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Fire Rated Metal Composites

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AJMAN - U.A.E

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FACADE SYSTEM

DRG. TITLE:
BS-8414 PART-2 SYSTEM
DEVELOPMENT- ARFS
MATERIAL SPECS & INFORMATION

FIRE LAB:
الفطيم Al-Futtaim اليمنت element

Al-Futtaim اليمنت element
Al Futtaim Element Materials Technology Dubai L.L.C
Facade Testing & Advisory Services Division

Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1
Drawing Number				Rev
BS-8414-P2-2248				03

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FACADE SYSTEM

DRG. TITLE:
BS-8414 PART-2 SYSTEM
DEVELOPMENT- ARFS
MATERIAL SAMPLE BOARD

FIRE LAB.:
الفطيم Al-Futtaim
اليمنت element

Scale	Drawn	Checked	Approved	Size
AS SHOWN	MJ	RAM	MUSHTAQ	A1

Drawing Number	Rev
BS-8414-P2-2249	00

الفطيم Al-Futtaim اليمنت element
Al Futtaim Element Materials Technology Dubai L.L.C
Facade Testing & Advisory Services Division

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-End of report-